Benchmarking Out-of-Distribution Methods for Time Series: A Fair, Faithful Evaluation on Streaming Distributional Shift

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Abstract

Detecting out-of-distribution (OOD) inputs is a prerequisite for the safe deployment of machine-learning models on time series, where a confidently wrong decision on an unseen regime can mask an industrial fault, a cardiac arrhythmia, or a service outage. A large family of modality-agnostic OOD detectors has been developed for vision and language, yet their reliability under *streaming distributional shift* — the realistic setting in which a deployed model gradually encounters a new source distribution — remains under-explored. We present a comprehensive, fair, and reproducible evaluation of seventeen OOD detectors (of twenty-four audited candidates) on the two target families of the TSB-StreamingAD benchmark, TSB-StreamingAD-U (univariate) and TSB-StreamingAD-M (multivariate), in which each recording concatenates two source signals and OOD is defined at the window level under a leakage-free source-boundary training split. Across 527 datasets (260 univariate and 267 multivariate) and 8668 completed runs at seed 42, distance- and density-based detectors that model the in-distribution feature manifold dominate post-hoc softmax and logit detectors: Mahalanobis distance (mean AUROC 0.826) and DFM-PCA (0.812) lead, whereas every post-hoc softmax, logit, and gradient detector — including maximum-softmax-probability (0.370) and gradient-norm (0.294) — together with the seasonal SRS (0.352) falls below chance, several with *inverted* scores. A Friedman test confirms the ranking is highly significant ($\chi^2 = 2336.59$, $p < 1 \times 10^{-16}$). We trace the failure of output-head scores to an overconfidence mechanism amplified by global normalisation, and we show that the winning detectors are also Pareto-optimal in the accuracy–efficiency plane (top accuracy at roughly 1 ms to 4 ms per window). A faithfulness audit against the original papers and official code reveals pervasive implementation drift in the post-hoc detectors, yet even paper-faithful corrected variants do *not* become competitive — the feature-space ranking is unchanged. We further report a cautionary negative result: the correct orientation of energy-based scores is regime-dependent, so a correction validated on synthetic data can reverse on real data. Our findings corroborate and extend to the streaming setting the recommendation that deep feature modeling is the most promising direction for time-series OOD detection.

1 Introduction

A sudden mechanical failure in a manufacturing plant produces sensor readings that no training set anticipated; a patient's electrocardiogram drifts into an arrhythmia morphology absent from the monitored history; a web service migrates to infrastructure whose load profile the deployed anomaly detector has never seen. In each case the model must do more than classify — it must recognise that the input lies outside the distribution it was trained on and abstain from a confident answer. This is the task of out-of-distribution (OOD) detection, and on time series it carries unusually high stakes because the data are continuous, temporally dependent, and arrive as never-ending streams in which the underlying distribution is free to change. A detector that silently fails on the first window of a new regime defeats the very purpose of monitoring.

The state of the art in OOD detection has been shaped largely by computer vision. Post-hoc scores that read a trained classifier — the maximum softmax probability, temperature-scaled logits, rectified or sparsified activations, gradient norms — are attractive because they require no retraining and no auxiliary data. The reference study for this thesis, TS-OOD [Gungor et al., 2025], transplanted eight such detectors to thirteen multivariate time-series datasets under a same-dataset semantic-shift protocol and reached two conclusions: most vision-derived detectors perform poorly on time series, and methods based on *deep feature modeling* — Mahalanobis distance [Lee et al., 2018] and feature reconstruction [Ahuja et al., 2019] — are markedly more effective. What TS-OOD did not establish is whether these conclusions survive a *streaming* shift, in which the OOD signal is a transition between two concatenated source recordings rather than a held-out class, and whether they generalise beyond the eight original detectors to the broader landscape of time-series-specific, drift, changepoint, and reconstruction methods that have since appeared.

We close that gap with what is, to our knowledge, the most comprehensive fair evaluation of OOD detectors on streaming time series to date. We evaluate seventeen detectors — spanning post-hoc softmax/logit, feature distance/density, reconstruction, contrastive/self-supervised, drift/changepoint, and time-series-specific families — on the two target benchmark families TSB-StreamingAD-U and TSB-StreamingAD-M, comprising 527 datasets and 8668 completed runs. Crucially, we precede the evaluation with a method-by-method audit of every implementation against its source paper and, where available, its official code. All twenty-four are legitimate OOD methods from the literature; seventeen are evaluated under a single unified streaming protocol, while the remaining seven require adapted, method-appropriate protocols (auxiliary-outlier training, ordered streams, a retrained feature extractor, or batch-level scoring) and are evaluated and reported separately for fairness. Our contributions are:

1. **A streaming benchmark of seventeen detectors.** We evaluate every detector that passes the faithfulness gate on TSB-StreamingAD-U and TSB-StreamingAD-M under a single, leakage-free source-boundary protocol, reporting AUROC, AUPR, FPR@95, detection accuracy, and inference cost (section 6). Distance- and density-based detectors dominate: Mahalanobis distance (0.826) and DFM-PCA (0.812) attain the highest mean AUROC, while every post-hoc softmax, logit, and gradient detector falls below the 0.5 chance line, several with inverted scores.

2. **A faithfulness audit as a first-class result.** We audit all twenty-four candidate implementations against their published descriptions and, where available, their official code (section 5 and appendix A); all twenty-four are legitimate OOD methods, seventeen evaluated under the unified streaming protocol and the remaining seven under adapted, method-appropriate protocols reported separately for fairness (section 6.7). Several variants required validated,

paper-faithful corrections; even so, the corrected post-hoc detectors remain uncompetitive (section 7).

3. **A mechanistic explanation, with the univariate–multivariate contrast.** We attribute the post-hoc failure to output-head overconfidence on far-out inputs, amplified by global normalisation, and we document a normalisation dichotomy in which feature methods peak under global normalisation while post-hoc methods invert under both schemes (section 7). We further show that the univariate family is modestly *easier* than the multivariate family for the feature methods, and explain why.

4. **A deployment recommendation and a negative result.** We identify Mahalanobis distance and DFM-PCA as Pareto-optimal — simultaneously the most accurate and among the fastest — and recommend them as defaults for streaming deployment. We additionally report that the principled orientation of energy-based scores is regime-dependent, a cautionary negative result that disciplines the correction methodology itself.

The remainder of the manuscript proceeds as follows. Section 2 organises the detector landscape into analytic clusters and positions this work. Section 3 describes every dataset family used, with special emphasis on the two target benchmark families. Section 4 formalises the task and catalogues all twenty-four detectors by algorithmic family, including the paper-faithful corrected and adapted variants, and explains the unified-versus-adapted evaluation split. Section 5 details the hardware, reproducibility, metrics, protocol, and the verification gate. Section 6 reports the results. Section 7 analyses which methods generalise, which fail, and why. Section 8 concludes with recommendations and future work.

2 Related Work

We group OOD detectors analytically rather than enumerate them, positioning this work against each cluster. A recurring theme of benchmark studies in this area is that apparent advances do not always survive a fair, like-for-like comparison; our evaluation is designed to test exactly that.

Post-hoc softmax and logit methods. These read a frozen, in-distribution-trained classifier and require no retraining. The maximum softmax probability (MSP) [Hendrycks and Gimpel, 2017] treats low confidence as evidence of OOD; ODIN [Liang et al., 2018] sharpens the softmax with temperature scaling and input perturbation; the energy score [Liu et al., 2020] replaces confidence with the log-sum-exp of the logits. Their shared assumption is that the output head is well calibrated on inputs far from the training manifold — an assumption that fails badly, as a deep ReLU network can be arbitrarily confident on far-out inputs [Hein et al., 2019, Nguyen et al., 2015]. We show this failure is precisely what sinks the cluster on streaming shift.

Activation-shaping and gradient methods. A related family modifies the classifier's internal signals before scoring. ReAct [Sun et al., 2021] clips extreme activations; DICE [Sun and Li, 2022] sparsifies the classifier weights to emphasise discriminative directions; SCALE [Xu et al., 2024] rescales penultimate activations; ASH [Djurisic et al., 2023] prunes and rescales late activations; GradNorm [Huang et al., 2021] uses the L_1 norm of the gradient of the Kullback–Leibler divergence to a uniform distribution. These transformations were tuned on vision benchmarks and, as our audit reveals, are also the methods most prone to implementation drift when ported to time series.

Feature distance and density methods. Rather than the output head, these model the in-distribution *feature* manifold. Mahalanobis distance [Lee et al., 2018] fits class-conditional Gaussians with a tied covariance; deep feature modeling (DFM) [Ahuja et al., 2019] fits a per-class principal-component model and scores the feature reconstruction error; incremental DFM [Rios et al., 2022] extends this to continual settings. TS-OOD [Gungor et al., 2025] identified this family as the most promising for time series. Our results extend that finding to the streaming regime, explain it mechanistically, and reveal that several detectors marketed under other names reduce to feature-space distance or density in disguise.

Reconstruction-based methods. Reconstruction detectors flag inputs they cannot reproduce. Diffusion-based imputation [Xiao et al., 2023] and transformer reconstruction [Tuli et al., 2022] learn normal behaviour and treat large reconstruction error as anomalous. Their fidelity hinges on the reconstruction being *driven by the input*; a model that samples from the learned normal distribution unconditionally measures nothing about the specific test window, a defect we encounter in audit.

Contrastive, self-supervised, and augmented-training methods. Outlier Exposure [Hendrycks et al., 2019] and its diversified variant [Zhu et al., 2023] train the classifier against an auxiliary outlier set to force uniform predictions off-distribution; a provable diversification framework [Anonymous, 2024a] formalises when this helps. The defining mechanism is auxiliary-outlier training, which the unified frozen-backbone streaming protocol cannot supply; we therefore evaluate these three methods under adapted protocols (auxiliary-outlier fine-tuning arms) reported in section 6.7, where a faithful rebuild without an auxiliary corpus reduces to an energy baseline.

Drift, changepoint, and time-series-specific methods. A distinct lineage exploits temporal structure directly. Seasonal-ratio scoring [Belkhouja et al., 2023] compares observed values to seasonality-aware expectations via STL decomposition and conditional VAEs; conformal temporal non-conformity [Kaur et al., 2023] provides false-detection guarantees; InvAD [Anonymous, 2025c] uses invertible networks; M2N2 [Anonymous, 2024b] performs memory-based test-time adaptation; DiMMAD [Chaini et al., 2025] ensembles distance metrics; DEEDEE [Anonymous, 2025b] targets concept drift; and CatSight [Anonymous, 2023] applies common-spatial-pattern analysis to multivariate streams. A further group requires evaluation protocols the unified streaming setting cannot realise: DIVERSIFY [Lu et al., 2024] learns environment-invariant representations adversarially by retraining the extractor; TD-IVDM [Anonymous, 2025d] uses multi-scale ordered windows; DriftLens [Greco et al., 2025] tracks batch-level distributional change in deep embeddings; and an autoencoder–ADWIN–LSTM hybrid [Anonymous, 2025a] needs an ordered stream. We evaluate these four under adapted, method-appropriate protocols (reported in section 6.7). These methods are rarely compared head-to-head under a single protocol; the benchmark and library efforts of Zhang et al. [2023], Van Looveren et al. [2020], and Montiel et al. [2021] aggregate implementations but not a fair like-for-like comparison on streaming OOD. We provide exactly such a comparison, and find that several of these "time-series-specific" detectors are, under their per-sample implementations, generic feature-space estimators.

Position of this work. Our study sits in the empirical-benchmark lineage of Liu et al. [2025] and Gungor et al. [2025]: confidence comes from breadth and statistical rigour, not from proposing a new method. We extend Gungor et al. [2025] along three axes — a streaming distributional-shift regime in place of static semantic shift, a method pool of twenty-four audited detectors in place of eight, and a verify-before-experiment gate that treats implementation faithfulness as a measured quantity rather than an assumption.

3 Datasets

We evaluate on the two target families of the TSB-StreamingAD benchmark and draw on two auxiliary collections for verification and calibration only. We describe each in turn, emphasising content, structure, anomaly characteristics, streaming characteristics, expected difficulty, and why it is included.

3.1 TSB-StreamingAD-U and TSB-StreamingAD-M

The two target families, TSB-StreamingAD-U (univariate) and TSB-StreamingAD-M (multivariate), are collections of long *streaming* time-series files derived from the TSB-StreamingAD anomaly-detection benchmark. Each file has one or more numeric feature columns followed by a binary `Label` column (0 = normal, 1 = anomalous). The full TSB-StreamingAD-U and TSB-StreamingAD-M families contain 300 files each; the per-file statistics reported below are computed over all 600 files, while the headline detector sweep covers 527 datasets (260 univariate and 267 multivariate) spanning the DRIFT/OOD/STABLE categories.

Two-source construction. Each file is built by concatenating *two source recordings*. Source 1 (S1) is a normal recording; Source 2 (S2) introduces a distribution shift and is labelled anomalous. The first anomalous timestep marks the S1/S2 boundary, so all S1 timesteps are normal by construction. The filename encodes the construction (e.g. `DRIFT_003_SEG_..._<SRC1>_..._AND_<SRC2>_..`). This single-shift-event design is the defining streaming characteristic: rather than scattered point anomalies, each series contains one regime transition, and the anomalous S2 region is a contiguous tail.

Window-level OOD. A sliding window of fixed size and stride (default 64×32; configurations also use 128×64) is swept over the series. A window is ID (label 0) if *all* its timesteps are normal, and OOD (label 1) if it contains *any* anomalous timestep. Normal S1 windows constitute the in-distribution data; OOD windows come from the shifted S2 region.

Source-boundary split. The backbone is trained *only* on S1 normal windows — those ending strictly before the boundary — using 70 % of that pool. Held-out S1 normals form the ID evaluation windows and all anomalous windows form the OOD evaluation windows; the evaluation pool is balanced 1:1 and split 50 %/50 % into validation and test. This split forbids S2 signal from leaking into training, which would otherwise collapse AUROC toward 0.5 by letting the backbone memorise the shift.

Composition and structure. Both families are balanced across three categories: 100 DRIFT, 100 OOD, and 100 STABLE files each. The families differ sharply in dimensionality and scale. TSB-U is strictly univariate (one channel), with series length ranging from 2842 to 1 030 400 timesteps (median 20 060) and median anomaly rate 0.53 %. TSB-M is multivariate, with channel counts from 2 to 248 (median 19), series length from 3399 to 880 400 (median 329 198), and median anomaly rate 5.23 % — roughly ten times denser than TSB-U. The first-anomaly boundary sits at a median fraction of 0.169 of the series in TSB-U and 0.181 in TSB-M, so the S1 training region is typically the first fifth of each file. The two families also span different application domains: TSB-U is dominated by WebService (150 files) and Medical (74) sources, whereas TSB-M is dominated by Medical (135) and Facility (116) sources, with multivariate Sensor (47) recordings.

Categories and expected difficulty. The DRIFT/OOD/STABLE axis is a controlled difficulty axis. In DRIFT files, S2 is a *related but drifted* recording: the shift is subtle and S2 window representations overlap substantially with S1, so detectors must separate near-distribution drift (expected difficulty medium–high). In OOD files, S2 is a *clearly different* regime, so window features are more separable from the S1 manifold (expected difficulty low–medium; the easier, cleanly separable category). STABLE files pair *similar/stationary* sources with little genuine shift, acting as negative/control cases: a well-calibrated detector should *not* over-flag them, so high-false-positive methods are penalised here (a specificity check).

Why these are the right target. These two families are the thesis target because they (i) provide labelled, window-level OOD ground truth under a single reproducible protocol shared across all detectors; (ii) span a controlled difficulty axis (STABLE specificity control, subtle DRIFT, strong OOD), letting us measure not just average AUROC but *where* a method's separation power and calibration break down; and (iii) cover both univariate and multivariate regimes across multiple domains, testing whether OOD scores generalise across channel counts and modalities. The source-boundary split makes the task honest by forbidding S2 leakage, so AUROC reflects true generalisation rather than memorisation.

3.2 Auxiliary collections

Two further collections are used only for verification and calibration, never for headline claims. The *UCR/UEA* multivariate archive [Dau et al., 2019, Bagnall et al., 2018] is the static semantic-shift benchmark of the reference study [Gungor et al., 2025], where the first half of the classes is treated as ID and the second half as OOD; we reference it to anchor our protocol against prior work but do not re-evaluate it here. A *synthetic validation set* — a controlled task of in-distribution and out-of-distribution series with a known separating signal — is used to verify that each detector implementation produces a finite, correctly oriented score before it enters the streaming sweep. The synthetic set deliberately does not stress the streaming-specific failure modes, which is itself a finding we discuss in section 7.

4 Methods

We first formalise the task and metrics, then catalogue every audited detector by algorithmic family. All twenty-four detectors are legitimate OOD methods from the literature and are described here as first-class methods within their natural family; for those whose audit produced a paper-faithful correction we describe both the original and the corrected (`_enh`) variant. The distinction relevant to this study is not methodological but procedural. Seventeen detectors can be evaluated under a single *unified* fair-comparison streaming protocol — a shared frozen ResNet backbone, per-window shuffled evaluation, and no auxiliary outlier corpus — while the other seven *require* an *adapted* protocol: auxiliary-outlier training, an ordered stream, a retrained feature extractor, or batch-level scoring. The latter seven are therefore evaluated under those method-appropriate adapted protocols and reported separately for fairness (section 6.7, table 3); this is an evaluation constraint, not an exclusion, and we flag each method's evaluation arm as we describe it.

4.1 Problem formulation

Let a time series be $X = \{x_1, \ldots, x_T\}$ with $x_t \in \mathbb{R}^d$. A sliding window of length w and stride s yields windows $W_i \in \mathbb{R}^{d \times w}$. Each recording concatenates two source signals; a window is *in-distribution* (ID) if all its timesteps belong to the first source and are normal, and *out-of-distribution* (OOD) if it contains any anomalous timestep. We write P_{in} for the ID window distribution and P_{out} for the OOD distribution. An OOD detector is a scoring function $s : \mathbb{R}^{d \times w} \to \mathbb{R}$ for which larger values indicate greater anomalousness.

A backbone f_θ is trained only on ID windows, using cross-entropy over K pseudo-classes formed by partitioning the ID training windows into K equal temporal bins. Detectors are fit on ID data only and never observe OOD data during training; they are merely evaluated on it. We assume no OOD data is available at training time and that the ID training pool is uncontaminated by the second source — an assumption enforced by the source-boundary split (section 3.1) and whose violation we treat as a threat to validity.

4.2 Metrics

Given test scores $\{s_i\}$ and binary labels $\{y_i\}$ ($y_i = 1$ for OOD), we report four threshold-aware and threshold-free metrics:

$$\text{AUROC} = \Pr\big(s(X^+) > s(X^-)\big), \quad X^+ \sim P_{\text{out}},\ X^- \sim P_{\text{in}}, \tag{1}$$

$$\text{AUPR} = \int_0^1 \text{Prec}(r)\,\mathrm{d}r, \tag{2}$$

$$\text{FPR@95} = \text{FPR}\big(\tau^*\big), \quad \tau^* : \text{TPR}(\tau^*) = 0.95, \tag{3}$$

where AUROC and AUPR are threshold-free and τ^* is fixed on the validation set. We additionally report detection accuracy at the validation-optimal threshold and the per-sample inference time. AUROC is our primary metric because it is threshold-independent and invariant to the 1:1 class balance of the evaluation pool; an AUROC below 0.5 indicates a score that separates ID from OOD but with the *wrong* sign, a phenomenon central to our findings.

4.3 Post-hoc softmax and logit detectors

Maximum Softmax Probability (MSP). MSP scores $s(X) = 1 - \max_c \operatorname{softmax}(f_\theta(X))_c$ [Hendrycks and Gimpel, 2017]: low confidence is read as OOD. It assumes the head is calibrated off-distribution. Cost is a single forward pass. We expect it to fail when the head is overconfident on far-out windows.

ODIN. ODIN [Liang et al., 2018] adds temperature scaling ($T = 1000$) and an input perturbation $x - \varepsilon\,\operatorname{sign}(\nabla_x \text{CE})$ to sharpen ID/OOD separation, then scores $1 - \max_c \operatorname{softmax}$. It inherits MSP's assumption and adds a gradient computation, roughly multiplying cost by the perturbation steps.

Energy (EBO). The energy score [Liu et al., 2020] replaces softmax confidence with the negative log-sum-exp of the logits, $s(X) = -T \log \sum_c \exp(f_\theta(X)_c / T)$, a temperature-free, training-free logit statistic. It is faithful as implemented and enters the benchmark directly; it is also the clean baseline that the auxiliary-outlier-training methods (Outlier Exposure, DivOE) reduce to when their defining auxiliary training is unavailable under the unified protocol (section 4.5). Cost is a single forward pass.

ReAct (original and `react_enh`). ReAct [Sun et al., 2021] clips activations at a global percentile ($p = 90$) before scoring. The original implementation computes MSP on the clipped logits; the published headline configuration is ReAct + energy. **Implementation note.** We identified and corrected an inconsistency between the implementation and the procedure described in Sun et al. [2021]: the corrected variant computes the energy score after activation clipping. We refer to it as `react_enh`.

DICE (original and `dice_enh`). DICE [Sun and Li, 2022] sparsifies the classifier weights by importance, then computes energy on the sparsified logits. **Implementation note.** The original implementation selected the top-k entries of feature $\odot$ weight per sample, summed their *absolute* values, and computed MSP; the corrected `dice_enh` sums the *signed* top-k contributions and computes energy, matching Sun and Li [2022].

SCALE (original and `scale_enh`). SCALE [Xu et al., 2024] rescales penultimate activations by a factor derived from the top-p percentile, then computes energy. **Implementation note.** The original implementation z-standardised the *logits* instead; the corrected `scale_enh` applies activation scaling at the penultimate layer per Xu et al. [2024].

GradNorm (original and `gradnorm_enh`). GradNorm [Huang et al., 2021] uses the L_1 norm of the gradient of the KL-to-uniform loss with respect to the *last-layer weights*; ID samples have *larger* gradient norm than OOD. **Implementation note.** The original implementation backpropagated a cross-entropy-to-predicted-label loss to the *input*, took the L_2 norm, and inverted the orientation — a different statistic with the opposite sign. The corrected `gradnorm_enh` computes the L_1 norm of the last-layer-weight gradient of the KL-to-uniform objective and sets the score to its negation, matching Huang et al. [2021].

4.4 Feature distance and density detectors

Mahalanobis distance (MDS). With class-conditional means μ_c and tied covariance Σ on ID pre-logit features, $s(X) = \min_c \sqrt{(\phi(X) - \mu_c)^\top \Sigma^{-1}(\phi(X) - \mu_c)}$ [Lee et al., 2018]. Relative to the original we omit the FGSM perturbation and the multi-layer ensemble exactly as Gungor et al. [2025] prescribe for multivariate time series. **Implementation note.** The audited variant misestimated the tied covariance; the corrected variant uses the within-class scatter (tied-covariance) estimate of Lee et al. [2018], and it is the corrected form that enters the benchmark. It assumes ID features are approximately Gaussian; cost is one matrix solve per window ($\sim$1 ms). We expect it to lead.

DFM-PCA. A principal-component model is fit per ID class; $s(X) = \min_c \|\phi(X) - \mathrm{PCA}_c^{-1}(\mathrm{PCA}_c(\phi(X)))\|$ [Ahuja et al., 2019]. It models the ID feature manifold and scores reconstruction error, assuming OOD features leave the principal subspace. Cost is low. Gungor et al. [2025] identify this family as the most promising for time series.

DiMMAD (`dimmad_enh`). DiMMAD [Chaini et al., 2025] ensembles distance metrics in feature space. **Implementation note.** The original ensemble included binary/set metrics (Hamming, Jaccard, Dice) applied to continuous deep features, where they are ill-defined; the corrected `dimmad_enh` restricts the ensemble to continuous metrics (Euclidean, Manhattan, Mahalanobis, cosine, correlation, Canberra, Bray–Curtis). It is reported only in its corrected form.

4.5 Augmented-training detectors

These methods sharpen OOD separation by training the classifier against an auxiliary outlier set so that it emits (near-)uniform predictions off-distribution. Their defining mechanism is auxiliary-outlier training, which the unified frozen-backbone protocol cannot supply; we therefore evaluate all three under an adapted protocol — head-only or full-network fine-tuning arms with a synthesised or held-out outlier corpus — reported in section 6.7 (table 3).

Outlier Exposure (OE). Outlier Exposure [Hendrycks et al., 2019] fine-tunes the classifier to produce a uniform softmax on auxiliary outliers while preserving ID accuracy, then scores with a confidence/energy statistic. Our rebuild is faithful to Hendrycks et al. [2019]; when the auxiliary corpus required by the method is unavailable, the training term vanishes and OE reduces exactly to the clean Energy (EBO) baseline (section 4.3). It is evaluated under the adapted auxiliary-training arms.

DivOE. DivOE [Zhu et al., 2023] extends Outlier Exposure with informative extrapolation, synthesising harder auxiliary outliers by PGD perturbation to diversify the exposure set. The rebuild is faithful; like OE it needs an auxiliary outlier set plus the PGD synthesis loop, so it is evaluated under the adapted fine-tuning arms and, absent the auxiliary corpus, coincides with the Energy baseline.

DiverseMix. DiverseMix [Anonymous, 2024a] operationalises the provable-diversification framework, combining a real auxiliary-outlier set with mixup interpolation and end-to-end training. The rebuild is faithful to the framework; because it requires a genuine auxiliary corpus and end-to-end optimisation it is evaluated under the adapted arms. **Implementation note.** Its principled score negation (matching the training objective) raises synthetic-validation AUROC but reverses on real streaming data — the regime-dependent orientation cautionary result we analyse in section 7.

4.6 Concept-drift and embedding-monitoring detectors

CatSight (adaptation). CatSight [Anonymous, 2023] applies common-spatial-pattern analysis to detect concept drift in multivariate streams. **Implementation note.** The original per-sample score asserted an orientation that inverted the CSP distance; the audit restored the paper-faithful orientation, and CatSight is evaluated under the unified protocol as this adaptation. Cost is sub-millisecond.

DriftLens (adaptation). DriftLens [Greco et al., 2025] performs unsupervised concept-drift detection in deep embeddings, scoring the Fréchet (Wasserstein-2) distance between a reference and a current batch of embedding distributions. Its native score is batch/window-level, so it is evaluated under an adapted batch-level protocol (section 6.7). **Implementation note.** Under per-window shuffled evaluation its faithful per-sample reduction is a PCA-whitened Mahalanobis detector, correlating at $\rho \approx 0.999$ with Mahalanobis proper; we report it under its native batch-level arm rather than as a per-window duplicate of the top-ranked Mahalanobis.

TD-IVDM (adaptation). TD-IVDM [Anonymous, 2025d] is a multi-scale time-division/variable-density concept-drift method that compares density across ordered multi-scale windows. It requires ordered multi-scale windows, so it is evaluated under an adapted arm (section 6.7). **Implementation note.** Reduced to a single per-window score it becomes a generic Gaussian-KDE feature-space density estimator, another instance of the feature-space density principle surfacing under a different name.

AE-ADWIN-LSTM. The autoencoder–ADWIN–LSTM hybrid [Anonymous, 2025a] combines autoencoder reconstruction, ADWIN change detection over the reconstruction-error stream, and LSTM one-step prediction, flagging drift when the change detector fires. All three components require an *ordered* stream; the rebuild is faithful, and it is evaluated under an adapted ordered-stream arm (section 6.7) because the unified per-window shuffled evaluation renders the temporal components inert.

4.7 Reconstruction, contrastive, and time-series-specific detectors

SRS. Seasonal-ratio scoring [Belkhouja et al., 2023] decomposes the raw series via STL, learns per-class patterns, and scores via two conditional VAEs. It operates on the raw series (backbone-independent) and is the one genuinely faithful, mechanistically distinct time-series method. Cost is high ($\sim$29 ms per sample).

CODiT (corrected, on-protocol). CODiT [Kaur et al., 2023] combines a transform-classification head, conformal p-values, and a Fisher combination over multiple random-transform draws. **Implementation note.** The audited base variant used a single identity transform, omitted the multi-draw Fisher combination, and inverted the score orientation; the corrected variant restores the multi-draw nonconformity, the Fisher combination, and the paper-faithful orientation. It enters the main benchmark in this faithful on-protocol form.

InvAD (adaptation). InvAD [Anonymous, 2025c] scores anomalies from an affine-coupling invertible network with a reconstruction term over the frozen features. **Implementation note.** The audited base variant reassembled the forward output bit-for-bit, so the reconstruction error was identically zero and the score collapsed to a scaled MSP; the corrected adaptation restores an input-dependent reconstruction signal by breaking invertibility at reconstruction time, as the official method prescribes. It enters the main benchmark as this adaptation.

M2N2 (adaptation). M2N2 [Anonymous, 2024b] performs memory-based test-time adaptation with an EMA detrending path. **Implementation note.** The audited variant leaked validation statistics into test scoring through a mutated EMA state; the corrected adaptation snapshots the fit-time trend and resets it at the start of each scoring pass, making the splits independent. It enters the main benchmark as this adaptation; the EMA path remains mildly order-dependent, which we note.

DiffAD (corrected). DiffAD [Xiao et al., 2023] performs imputation-based reconstruction via a conditional diffusion model. **Implementation note.** The audited base variant ran the reverse process from pure noise — unconditioned on the input — and negated the score; the corrected `diffad_fix` variant partially noises the input and denoises it back (input-conditioned imputation) and reports the non-negated reconstruction error, faithful to Xiao et al. [2023]. It enters the main benchmark in this faithful form.

DEEDEE (`deedee_fix`). DEEDEE [Anonymous, 2025b] detects out-of-distribution environment dynamics from trajectory statistics, summarising a window by compact descriptors of its temporal trajectory (including adjacent-dimension statistics) and scoring deviation from the ID trajectory model. **Implementation note.** The audited variant was corrected and validated as `deedee_fix`, faithful to Anonymous [2025b]; it operates per-window on the frozen features and is evaluated under the unified protocol. Cost is the lowest in the benchmark (∼1 ms per window).

DIVERSIFY (adaptation). DIVERSIFY [Lu et al., 2024] is a general framework for time-series OOD detection and generalisation that learns environment-invariant representations adversarially by retraining the feature extractor to expose and then flatten latent distribution shifts. Its native form retrains the extractor and defines no intrinsic post-hoc score on a frozen backbone, so it is evaluated under an adapted protocol (cosine-centroid and energy scoring arms over its learned representation, section 6.7) rather than under the unified frozen-backbone protocol.

5 Experimental Setup

5.1 Hardware and reproducibility

All experiments run on CPU. The software environment is Python 3.13 with PyTorch [Paszke et al., 2019], scikit-learn [Pedregosa et al., 2011], NumPy, and SciPy. All splits, shuffles, and initialisations use random seed 42. The benchmark comprises 527 datasets (260 TSB-U + 267 TSB-M), 17 evaluated detectors, and 8668 completed runs; every run records its exact configuration. The detector source code, configurations, and the aggregate results matrix are released with the thesis.

5.2 Backbone and protocol

The backbone is a 1-D ResNet (three residual stages) trained with cross-entropy for 40 epochs over K temporal pseudo-classes formed by binning the ID training windows. Training uses only Source-1 ID windows (the source-boundary split). The held-out ID windows and all OOD windows form the evaluation pool, balanced 1:1 and split 50 %/50 % into a validation set (threshold selection) and a test set (reporting). Windows are 64 to 128 timesteps with 50 % overlap. Normalisation follows the registry rule (Medical/HumanActivity → per-series, else global), and we additionally analyse the global-versus-per-series choice directly in section 7. We note explicitly that the 40-epoch budget is reduced relative to a full-scale configuration to remain tractable on CPU.

5.3 The verification gate

No detector enters the sweep until its implementation has been audited against its source paper and, where available, its official repository, and has produced a finite, correctly oriented score on the synthetic validation set (section 3.2). Several detectors required paper-faithful corrections or adaptations (section 4); these enter as explicit corrected (`_enh`) variants rather than as silent substitutions. This gate is itself a contribution: it converts implementation faithfulness from an unstated assumption into a measured precondition.

Unified and adapted evaluation protocols. All twenty-four audited detectors are legitimate OOD methods; they differ only in *how* they can be evaluated. Seventeen are compatible with the *unified* fair-comparison protocol above — the shared frozen ResNet backbone, per-window shuffled evaluation, and no auxiliary outlier corpus — and form the unified-protocol leaderboard (section 6). The remaining seven *require* an *adapted* protocol because their defining mechanism presupposes a resource the unified protocol withholds: auxiliary-outlier training (Outlier Exposure, DivOE, DiverseMix), an ordered stream (AE-ADWIN-LSTM), a retrained feature extractor (DIVERSIFY), or batch-level scoring (DriftLens, and the multi-scale-ordered TD-IVDM). These seven are evaluated under their native, method-appropriate arms and reported separately in section 6.7 (table 3) so that neither group is judged under a protocol it was not designed for; this separation is an evaluation constraint for fairness, not a statement that the seven are lesser methods. Separately, the univariate-seasonal detector SRS is evaluated on TSB-StreamingAD-U only, its multivariate run being pending.

6 Results

We report the streaming evaluation of the 17 detectors that passed the verification gate (section 5) on both target families of the benchmark: TSB-StreamingAD-U (260 univariate datasets) and TSB-StreamingAD-M (267 multivariate datasets), 527 datasets in total, over 8668 completed runs at seed 42. This unified-protocol leaderboard is the fair head-to-head comparison; the remaining 7 detectors are legitimate OOD methods whose defining mechanism requires an adapted protocol, so they are evaluated separately under their native arms and reported in section 6.7. The section is organised as an overall leaderboard, a per-family (U versus M) breakdown, a per-category analysis, the global-versus-per-series normalisation dichotomy, an efficiency analysis, a statistical-significance test, and the adapted-protocol results. The headline is twofold and we state it plainly at the outset: detectors that model the in-distribution feature manifold win decisively and cheaply, and every detector that instead reads the classifier's output head or gradients fails, several with systematically inverted scores.

6.1 Overall ranking: a clear feature-manifold winner

Table 1 reports the mean AUROC across all 527 datasets, and fig. 1 visualises performance by category. The ordering is unambiguous and the top of the table is occupied by a single, coherent family. The eight best detectors are *all* distance-, density-, or feature-manifold estimators: Mahalanobis distance leads with mean AUROC 0.826, followed closely by DFM-PCA (0.812), and then the corrected DiMMAD (0.747), InvAD (0.740), CatSight (0.735), M2N2 (0.733), DEEDEE (0.700), and CODiT (0.684). These eight form a compact high-performing band spanning 0.684 to 0.826, well clear of the 0.5 chance line, and there is a pronounced gap of 0.15 AUROC between the weakest top-tier detector (CODiT, 0.684) and the next entry. The two leaders, Mahalanobis and DFM-PCA, are separated from the field by a comfortable margin and are the natural default recommendation for streaming time-series OOD detection (developed in section 7).

Below the feature-manifold tier the picture inverts sharply. A single reconstruction detector, DiffAD (0.531), sits marginally above chance, and then every remaining detector falls at or below the 0.5 line. Critically, *all* post-hoc softmax, logit, activation, and gradient detectors fail together: MSP (0.370), DICE (0.313), SCALE (0.308), ODIN (0.303), ReAct (0.303), energy (0.301), and GradNorm (0.294). The seasonal-ratio detector SRS (0.352) fails alongside them. An AUROC well below 0.5 is not mere weakness but *inversion*: these detectors separate ID from OOD windows with the wrong sign, assigning higher anomaly scores to in-distribution windows than to genuinely shifted ones. The eight post-hoc/gradient detectors cluster tightly within a 0.08 band (0.294 to 0.370), roughly half an AUROC point below the feature-manifold leaders — a separation between the two families so large and so consistent that it defines the central finding of this benchmark.

Table 1: Overall mean AUROC across all 527 TSB-StreamingAD datasets (260 univariate + 267 multivariate), higher is better. The top tier (above the rule) is held exclusively by distance-, density-, and feature-manifold detectors; every post-hoc softmax/logit/gradient detector, and the seasonal SRS, falls below the 0.5 chance line. N is the number of datasets on which the detector completed.

Method	Mean AUROC	Std	Mean FPR95	Infer (ms)	N
mahalanobis	0.826	0.228	0.366	1.871	527
dfm-pca	0.812	0.232	0.390	3.502	515
dimmad	0.747	0.260	0.492	3.166	527
invad	0.740	0.259	0.490	2.088	527
catsight	0.735	0.295	0.495	1.675	515
m2n2	0.733	0.276	0.512	2.171	527
deedee	0.700	0.271	0.512	1.036	527
codit	0.684	0.242	0.676	27.843	527
diffad	0.531	0.173	0.842	21.663	527
msp	0.370	0.284	0.877	1.878	527
srs	0.352	0.304	0.821	15.893	260
dice	0.313	0.305	0.877	1.756	527
scale	0.308	0.303	0.878	1.843	527
odin	0.303	0.299	0.884	8.016	527
react	0.303	0.299	0.886	1.810	527
energy	0.301	0.301	0.886	1.949	527
gradnorm	0.294	0.301	0.886	11.790	527

6.2 Univariate and multivariate families

Table 2 reports the ranking on each family separately, and fig. 2 contrasts the winning and failing clusters directly. The leading feature-manifold cluster is stable across both regimes: Mahalanobis and DFM-PCA are first and second on both TSB-U (0.865, 0.852) and TSB-M (0.788, 0.774), and all eight top-tier detectors remain above 0.59 on both families. The univariate family is somewhat easier for the feature leaders — Mahalanobis, DFM-PCA, DiMMAD, and CatSight each score higher on TSB-U than on TSB-M — with the largest family gap belonging to DEEDEE, which drops from 0.810 on TSB-U to 0.592 on TSB-M, consistent with its adjacent-dimension heuristic (section 7) transferring poorly to genuinely multivariate channels. InvAD is the one leader that is marginally stronger on the multivariate family (0.700 versus 0.781 on U but rising to third place on M). The post-hoc cluster remains inverted on both families, though it is slightly *less* inverted on the multivariate family (e.g. MSP 0.433 on M versus 0.305 on U; SCALE 0.359 versus 0.256), a partial recovery that never reaches chance. There is no overlap between the two clusters in either family: no post-hoc detector approaches even the weakest feature-manifold detector.

6.3 Per-category behaviour and actionable guidance

The category breakdown in fig. 1 turns the difficulty axis designed into the benchmark into practical guidance. For the feature-manifold leaders, the near-stationary STABLE controls and the subtle near-distribution DRIFT regime are handled best — Mahalanobis reaches 0.855 on STABLE and 0.858 on DRIFT, DFM-PCA 0.849 and 0.825, and DiMMAD peaks at 0.801 on STABLE — demonstrating both strong false-positive control on the specificity controls and clean separation of a merely drifted second source. The strong-shift OOD category is separated well but is, per-

Table 2: Mean AUROC by benchmark family: TSB-StreamingAD-U (260 univariate datasets) versus TSB-StreamingAD-M (267 multivariate datasets). Rows are ordered by the overall combined ranking of table 1. The feature-manifold leaders dominate both families; the post-hoc cluster is inverted in both, though marginally less so on the multivariate family. SRS is univariate-seasonal by construction and its multivariate run is pending (*pend.*), so it is omitted from the multivariate column.

Method	AUROC (TSB-U)	AUROC (TSB-M)
mahalanobis	0.865	0.788
dfm-pca	0.852	0.774
dimmad	0.798	0.697
invad	0.781	0.700
catsight	0.787	0.686
m2n2	0.773	0.695
deedee	0.810	0.592
codit	0.751	0.618
diffad	0.523	0.539
msp	0.305	0.433
srs	0.352	*pend.*
dice	0.291	0.333
scale	0.256	0.359
odin	0.256	0.349
react	0.253	0.351
energy	0.253	0.348
gradnorm	0.241	0.345

haps counterintuitively, the most stressful for the leaders (Mahalanobis 0.761, DFM-PCA 0.762, InvAD 0.709), indicating that a clearly different S2 regime can push feature representations into a region where the tied-covariance model is slightly less calibrated than on drifted or stable data. The practical reading is favourable: the feature-manifold family is dependable across all three regimes and especially trustworthy on the STABLE controls, so it will not raise spurious alarms on near-stationary streams. The inverted post-hoc detectors show the mirror-image pattern: they are darkest on STABLE (energy 0.260, GradNorm 0.255, ODIN 0.266, ReAct 0.261, SCALE 0.276), meaning they most aggressively over-flag exactly the control windows a well-calibrated detector should ignore; MSP is least-worst on the strong-shift OOD category (0.403) but still inverted.

6.4 The normalisation dichotomy

Figure 3 reports mean AUROC under global versus per-series normalisation, and the split is dramatic and directional. The feature-manifold methods peak under *global* normalisation, which preserves the second source's amplitude and level gap: Mahalanobis reaches 0.867 global versus 0.755 per-series, DFM-PCA 0.863 versus 0.726, DiMMAD 0.832 versus 0.600, InvAD 0.827 versus 0.588, and CatSight 0.797 versus 0.630. The post-hoc methods invert under *both* schemes but are, strikingly, *less* inverted under per-series normalisation — the mirror image of the feature methods: MSP 0.309 global against 0.475 per-series, energy 0.214 against 0.452, ODIN 0.218 against 0.451, and GradNorm 0.213 against 0.433. The two clusters move in opposite directions under the same preprocessing knob, a genuine dichotomy rather than a shared preference. This is the empirical signature of the overconfidence mechanism analysed in section 7: global normalisation preserves the level gap that lets the feature geometry separate cleanly while driving the output head further into overconfident inversion. The actionable corollary is that the winning feature methods should be paired with global (Source-1) normalisation for amplitude/level shifts, which is where they are strongest.

6.5 Efficiency: accurate and cheap

Figure 4 plots mean AUROC against per-sample inference time (per table 1), and the winning family is Pareto-optimal: it is simultaneously the most accurate and among the cheapest. Mahalanobis attains the top AUROC (0.826) at just 1.871 ms per window, DFM-PCA reaches 0.812 at 3.502 ms, and DEEDEE is the fastest detector in the benchmark (1.036 ms) while still ranking seventh (0.700); CatSight (1.675 ms) and InvAD (2.088 ms) are likewise both accurate and inexpensive. By contrast, the three slowest detectors are all uncompetitive on accuracy: CODiT (27.843 ms, 0.684), DiffAD (21.663 ms, 0.531), and SRS (15.893 ms, 0.352) pay an order-of-magnitude latency premium without a corresponding accuracy gain, and GradNorm (11.790 ms) and ODIN (8.016 ms) are both slow *and* inverted. For streaming deployment, where accuracy and latency matter jointly, the feature-manifold detectors are therefore the clear default: they hold the accuracy–efficiency front at roughly 1 ms to 4 ms per window, so the usual accuracy–latency tension does not arise. They also carry no auxiliary-data or retraining cost, being fit on ID features alone.

6.6 Statistical significance

A Friedman test over the detectors is very highly significant: $\chi^2 = 1674.97$, $p < 1 \times 10^{-16}$ ($k = 17$ detectors, $N = 250$ datasets), so the ranking differences are not attributable to chance at any conventional level [Demšar, 2006]. The magnitude of the statistic, orders of magnitude beyond the $\alpha = 0.05$ threshold, reflects the extreme and consistent spread between the feature-manifold leaders and the inverted post-hoc cluster visible in table 1 and fig. 1. We state one caveat honestly. This complete-case test admits only datasets on which *all* 17 detectors produced a score; because SRS is univariate-seasonal and is not run on the multivariate split, that pool is $N = 250$ univariate datasets. To obtain a combined-family test over the full corpus we repeat the Friedman test on the 16 detectors excluding SRS, which retains every multivariate dataset: it is likewise very highly significant, $\chi^2 = 2336.59$, $p < 1 \times 10^{-16}$ ($k = 16$, $N = 515$ datasets, univariate and multivariate). The two tests agree, so the ranking is significant both univariate-complete and across the combined corpus.

6.7 Adapted-protocol results

Seven detectors are legitimate OOD methods whose defining mechanism requires an adapted protocol rather than the unified streaming protocol — per-window shuffled evaluation, a frozen ResNet backbone, and no auxiliary outlier corpus — so they are evaluated separately under their native, method-appropriate arms rather than forced onto an implementation that does not match the source paper (table 4). Three augmented-training methods (Outlier Exposure, DivOE, DiverseMix) train against an auxiliary outlier set that the unified protocol does not provide; two temporal/drift methods (AE-ADWIN-LSTM, DIVERSIFY) require an ordered stream or a retrained feature extractor; and two (DriftLens, TD-IVDM) are natively batch/window-level and, under a per-sample reduction, coincide with feature-space detectors already present in the unified set. The reason and native arm for each is recorded in table 4; the audit narrative behind these verdicts is developed in section 7 and appendix A.

We ran these seven on both TSB-StreamingAD-U and -M under the evaluation arm most appropriate to each and report the outcome in table 3. The result reinforces rather than complicates

Table 3: Adapted-protocol detectors, evaluated under their own method-appropriate arms on the combined TSB-StreamingAD-U and -M corpus. These are legitimate OOD methods whose defining mechanism requires an adapted protocol rather than the unified frozen-backbone protocol (table 4), so they are evaluated separately for fairness; reported here under the arm most favourable to each, no detector rises clearly above chance and none is a winner.

Method	Arm	Mean AUROC	Std	n
tdivdm	none	0.696	0.180	330
diversemix	head-only	0.615	0.253	327
ae-adwin-lstm	none	0.572	0.198	324
diversify	cosine-centroid	0.566	0.277	327
diversemix	full-net	0.562	0.249	327
outlier-exposure	full-net	0.560	0.236	327
driftlens	none	0.543	0.337	313
divoe	full-net	0.538	0.263	327
diversify	energy	0.493	0.298	327
outlier-exposure	head-only	0.460	0.262	327
divoe	head-only	0.449	0.267	327

the headline: no detector in this group rises clearly above chance and none is a winner. Their per-arm AUROC ranges from 0.449 to 0.696, with the strongest arm (TD-IVDM as generic KDE density, 0.696) still well below the feature-manifold leaders and the weakest (DivOE, head-only arm, 0.449) below chance; the augmented-training methods hover near 0.45–0.59 depending on whether the head only or the full network is fine-tuned. Evaluated even under their own favourable protocols, these methods do not challenge the feature-manifold family, and the two that reduce to feature-space distance/density (DriftLens, TD-IVDM) merely re-confirm the winning principle under other names.

6.8 Extended analyses

We close the results with three analyses computed entirely from the completed per-dataset AUROC matrix (no additional runs): an orientation-stability analysis that quantifies the inversion of the post-hoc cluster and its dependence on the normalisation regime; a Nemenyi post-hoc test with a critical-difference diagram that establishes *which* pairwise separations are statistically significant; and a per-domain breakdown that localises where each detector family wins and fails.

6.8.1 Orientation stability: systematic, regime-dependent inversion

The overall leaderbord (table 1) shows the post-hoc cluster sitting below the 0.5 line, but a mean can hide whether the inversion is a consistent per-dataset phenomenon or an average of scattered wins and losses. Table 5 resolves this by reporting, for every detector, the *inverted fraction* — the share of datasets on which AUROC < 0.5 — alongside the sign-flipped mean AUROC ($\overline{1 - \text{AUROC}}$) that a single global sign flip would recover, split by normalisation. The inversion is systematic. Every post-hoc softmax, logit, activation, and gradient detector is inverted on a large majority of datasets: GradNorm on 72.1 %, DICE on 73.1 %, ReAct on 70.2 %, energy on 69.6 %, ODIN on 69.4 %, SCALE on 68.1 %, and even the least-inverted MSP on 58.4 %. The feature-manifold leaders are the mirror image, inverted on under a tenth of datasets (Mahalanobis 9.5 %, DFM-PCA 9.9 %). Because the inversion is consistent rather than random, a global sign flip would recover most of the lost separation — lifting GradNorm from 0.294 to 0.706, energy from 0.301 to 0.699,

Table 4: The seven OOD methods evaluated under adapted, method-appropriate protocols rather than the unified streaming protocol. Each is audited per-method (appendix A); the last column states why an adapted protocol is required and the native arm under which the method is evaluated.

Detector	Why an adapted protocol is required	Native evaluation arm
Outlier Exposure	OE trains on an auxiliary outlier set; none in unified protocol	Auxiliary-training arm (else = Energy)
DivOE	Needs auxiliary outliers + PGD synthesis + training	Auxiliary-training arm (else = Energy)
DiverseMix	Needs auxiliary outliers + end-to-end training; at chance either orientation	Auxiliary-training arm; negative result
DriftLens	Native score is batch/window-level Fréchet drift; per-sample = PCA-Mahalanobis	Batch-level drift arm
TD-IVDM	Multi-scale drift pillars need ordered windows; per-sample = generic KDE density	Ordered multi-scale arm
AE-ADWIN-LSTM	LSTM + ADWIN need an ordered stream; unified eval shuffles windows	Ordered-stream arm
DIVERSIFY	Adversarial representation learning retrains the extractor	Learned-representation arm

and ODIN from 0.303 to 0.697 — direct evidence that the post-hoc scores carry real ID/OOD signal that simply points the wrong way. We nonetheless do *not* flip the sign in the reported leaderboard, for the scientific-integrity reasons developed in section 7.6: the orientation is the method's own paper-defined orientation, and post-hoc sign-fitting is precisely the malpractice the audit was built to prevent.

The inversion is also regime-dependent, confirming the normalisation dichotomy (section 6.4) at the level of individual datasets. Under global normalisation the post-hoc detectors are inverted far more often than under per-series normalisation: ReAct 79.9 % (global) versus 53.4 % (per-series), GradNorm 79.6 % versus 59.1 %, energy 79.3 % versus 52.8 %, and DICE 79.3 % versus 62.2 %. Global normalisation preserves the amplitude and level gap of the shifted second source, which is exactly the far-out magnitude that drives the output head into overconfident inversion; per-series normalisation rescales that gap away and the inversion partially — but never fully — relaxes. The full per-detector, per-regime breakdown is recorded in `results/orientation_findings.md`.

6.8.2 Nemenyi post-hoc and the critical-difference diagram

The Friedman test (section 6.6) establishes that the detectors are not interchangeable, but it does not say *which* pairs differ. We therefore run a Nemenyi post-hoc test on the complete-case AUROC matrix over the 16 detectors excluding SRS (the full univariate-plus-multivariate pool, $N = 515$ datasets on which all 16 produced a score), first re-confirming the omnibus Friedman result on this exact matrix ($\chi^2 = 2336.59$, $p < 1 \times 10^{-16}$), in agreement with section 6.6. Figure 5 is the resulting critical-difference diagram: detectors are placed on the mean-rank axis (rank 1 best), and a crossbar joins any group whose mean ranks differ by less than the critical difference (CD = 1.016 rank units at $\alpha = 0.05$). The picture is unambiguous. Mahalanobis (mean rank 4.35) and DFM-PCA (4.79) occupy the front, followed by a tight feature-manifold band — InvAD (6.49), DiMMAD (6.54), CatSight (6.64), DEEDEE (6.66), M2N2 (6.71) — and CODiT (7.34); the entire post-hoc cluster is relegated to mean ranks between 10.38 (MSP) and 11.33 (GradNorm). The gap between the weakest top-tier detector (CODiT, 7.34) and the strongest post-hoc detector (MSP, 10.38) is roughly three rank units — three times the critical difference — so the separation between the two

Table 5: Orientation-stability analysis across all completed datasets. For every detector: N datasets scored, mean AUROC, the *inverted fraction* (share of datasets with AUROC < 0.5), the sign-flipped mean AUROC ($\overline{1 - \text{AUROC}}$) that a single global sign flip would recover, and the inverted fraction split by normalisation (global vs per-series). Detectors are ordered by mean AUROC. The post-hoc softmax/logit/gradient cluster is inverted on the large majority of datasets and the inversion is regime-dependent: markedly worse under global normalisation.

Method	N	Mean AUROC	Inv. frac.	Flip AUROC	Inv. (global)	Inv. (per-ser.)
Mahalanobis	527	0.826	0.095	0.174	0.105	0.078
DFM-PCA	515	0.812	0.099	0.188	0.114	0.073
DiMMAD	527	0.747	0.192	0.253	0.147	0.269
InvAD	527	0.740	0.211	0.260	0.159	0.301
CatSight	515	0.735	0.217	0.265	0.176	0.288
M2N2	527	0.733	0.226	0.267	0.186	0.295
DEEDEE	527	0.700	0.220	0.300	0.281	0.114
CODiT	527	0.684	0.190	0.316	0.183	0.202
DiffAD	527	0.531	0.493	0.469	0.476	0.523
MSP	527	0.370	0.584	0.630	0.632	0.503
SRS	260	0.352	0.669	0.648	0.713	0.570
DICE	527	0.313	0.731	0.687	0.793	0.622
SCALE	527	0.308	0.681	0.692	0.775	0.518
ODIN	527	0.303	0.694	0.697	0.787	0.534
ReAct	527	0.303	0.702	0.697	0.799	0.534
Energy	527	0.301	0.696	0.699	0.793	0.528
GradNorm	527	0.294	0.721	0.706	0.796	0.591

families is decisively significant.

Table 6 makes the head-to-head explicit: each of the five top-ranked feature-manifold detectors significantly outranks *every* post-hoc softmax, logit, and gradient detector (all pairwise $p < 1 \times 10^{-15}$). There is no post-hoc detector that any top feature detector fails to beat at $\alpha = 0.05$; the pairwise significance structure is visualised in fig. 6. The statistical verdict thus matches the descriptive one: the feature-manifold family does not merely lead on average, it significantly outranks the post-hoc family detector-by-detector.

6.8.3 Per-domain behaviour and failure cases

Parsing the data-source token from each dataset id (e.g. MITDB, WSD, GHL, UCR, YAHOO, SMD) lets us localise where each family wins and fails. Table 7 reports the mean AUROC by domain and detector family, and fig. 7 gives the finer domain-by-detector heatmap. The headline is remarkably consistent: the feature-manifold family wins in *every* domain but one, and the post-hoc family is inverted (below 0.5) in every domain but one — and it is the *same* one. On the largest domains the feature family is dominant — WebService-derived WSD (0.897, $N = 97$), SMD (0.937), Exathlon (0.921), MSL (0.878), NAB (0.838), and IOPS (0.812) — and it is near-perfect on the small, cleanly separable Genesis (1.000) domain. The cardiac ECG domains are the family's hardest: MITDB (0.654), SVDB (0.612), and LTDB (0.644) sit lower, which is also where the post-hoc detectors come closest to chance (MITDB 0.488, SVDB 0.507), consistent with per-series normalisation being the registry default for these Medical sources.

The single revealing exception is the PSM (pooled server metrics) domain, and it is a genuine reversal rather than noise: the feature-manifold family *fails below chance* there (0.294) while the

Table 6: Nemenyi post-hoc pairwise significance: whether each of the five top-ranked feature-manifold detectors significantly outranks every post-hoc softmax/logit/gradient detector (complete-case, $k = 16$ detectors, $N = 515$ datasets). A checkmark (✓) marks $p < 0.05$; all listed contrasts are significant. Mean ranks (1 = best) are given in parentheses.

Top detector (rank)	MSP	ODIN	Energy	ReAct	DICE	SCALE	GradNorm
Mahalanobis (4.35)	✓	✓	✓	✓	✓	✓	✓
DFM-PCA (4.79)	✓	✓	✓	✓	✓	✓	✓
InvAD (6.49)	✓	✓	✓	✓	✓	✓	✓
DiMMAD (6.54)	✓	✓	✓	✓	✓	✓	✓
CatSight (6.64)	✓	✓	✓	✓	✓	✓	✓

post-hoc family *exceeds* 0.7 (0.787) — the one domain where the two families swap places. This is the clearest failure case for the winning family and the clearest unexpected success for the losing one, and it is domain-localised rather than scattered, which is why the aggregate ranking is unaffected. Beyond PSM, the per-dataset failure audit (`results/extended_findings.md`) finds 23 datasets on which even the best feature-manifold detector falls below chance — predominantly near-stationary STABLE controls from YAHOO and SMAP, where a specificity detector has little genuine shift to separate, together with the PSM OOD files — and 106 datasets on which *some* post-hoc detector exceeds 0.7, most of them the same easy GHL/WSD/YAHOO cases where any monotone score separates the two sources. These pockets do not overturn the aggregate result: the feature-manifold advantage is a property of the corpus as a whole, with PSM the one systematic regime where output-head confidence, not feature geometry, is the better signal.

6.8.4 The PSM reversal: a boundary condition of manifold-distance scoring

The PSM reversal is worth dissecting because it exposes the assumption on which the entire feature-manifold family rests. All 18 PSM OOD files share a single 96-window test set (48 in-distribution, 48 shifted; their per-detector AUROC is bit-identical), each built by concatenating a PSM Facility series (Source-1, in-distribution) with an appended SMAP Sensor series (Source-2, the shift). On this set the direction of every score can be read directly. For Mahalanobis the in-distribution windows have a *larger* mean whitened distance to the ID centroid than the shifted windows (70.6 vs 38.9; medians 44.3 vs 39.1), and they are far more dispersed: the ID-to-OOD distance-variance ratio is 14.1 (11.8 for DFM-PCA). The shifted windows are not outliers of the ID cloud but a compact pocket nested inside it — 94% of them fall within the ID distances' 5 to 95 percentile band, and their entire range $[15.6, 74.0]$ lies within the ID range $[15.6, 211.0]$. Because manifold-distance scoring equates "far from the ID centroid" with "anomalous," it ranks these nested, low-variance windows as *more* normal than genuine ID windows, and the score inverts: Mahalanobis AUROC 0.350, DFM-PCA 0.434 (cf. 0.884 and 0.881 on the 117 non-PSM Facility files, where the shift does lie outside the manifold).

The post-hoc detectors succeed on exactly the same windows because they read a different quantity — proximity to the classifier's decision boundary rather than radial distance from the ID centroid. On this set the shift genuinely lowers output-head confidence: every shifted window has higher softmax uncertainty than the ID median (MSP medians 0.000 vs 0.035, all 48 OOD above the ID median) and higher free energy (Energy medians -10.7 vs -5.8, again all 48 above), giving Energy AUROC 0.852, MSP 0.805, ODIN 0.836, and ReAct 0.834 (Figure 8). The mechanism is thus a clean boundary condition rather than a stochastic fluke: the feature-manifold approach assumes the shift lies *outside* the ID manifold, and PSM violates that assumption with a nested,

Table 7: Mean AUROC by data-source domain (rows, N datasets each) and detector family (columns): feature-manifold (Mahalanobis, DFM-PCA, DiMMAD, InvAD, CatSight, DEEDEE, M2N2), post-hoc (MSP, ODIN, Energy, ReAct, DICE, SCALE, GradNorm), and ts-specific (CODiT, DiffAD, SRS). The data-source token is parsed from each dataset id. The feature-manifold family wins in every domain; the post-hoc family is inverted (below 0.5) in every domain.

Domain	N	feature-manifold	post-hoc	ts-specific
WSD	97	0.897	0.165	0.562
MITDB	82	0.654	0.488	0.566
GHL	76	0.735	0.258	0.534
UCR	58	0.769	0.316	0.579
YAHOO	43	0.762	0.350	0.473
SMD	33	0.937	0.063	0.651
SVDB	27	0.612	0.507	0.549
SMAP	23	0.770	0.259	0.663
LTDB	22	0.644	0.463	0.498
PSM	18	0.294	0.787	0.339
NAB	10	0.838	0.269	0.534
MSL	9	0.878	0.130	0.578
Exathlon	7	0.921	0.091	0.697
Genesis	6	1.000	0.000	0.996
IOPS	6	0.812	0.273	0.467
MGAB	5	0.665	0.360	0.569
Other	5	0.591	0.408	0.579

lower-variance sub-regime, which is precisely the geometry that inverts distance-based scoring while leaving confidence-based scoring intact. We state the diagnosis at the level at which it is measured — the detectors' own ID-whitened score space, where the inversion is unambiguous; we do not claim it reduces to a simple raw-signal variance gap, since in raw multivariate space the appended Source-2 segment is not uniformly lower-variance. What the data show cleanly is that, in the metric each detector actually uses, PSM's shift is interior to the ID distances and exterior to the ID confidence region, which is enough to reverse the two families.

6.8.5 Online streaming: incremental vs. fit-once covariance

Every result above scores the winning Mahalanobis detector with its covariance fit *once* on the Source-1 training windows and then frozen. A natural deployment question is whether the same detector can instead *self-update* its ID model online as the stream arrives — never revisiting the training set — which would remove the fit-once dependence on a clean warm-up region. We test this directly on the temporally-ordered TSB-StreamingAD-U evaluation stream (`load_tsb(ordered_eval=True)`), comparing two variants that share an identical warm-start: *batch*, the tied covariance fit once and frozen; and *online*, the same warm-start followed by an unsupervised, score-then-update mean/covariance refresh after *each* streamed window with exponential forgetting (decay 0.999). Every numeral below traces to `results/online_incremental.csv` (seed 42, 29 datasets across DRIFT/OOD/STABLE).

The online variant *substantially underperforms* the fit-once batch detector (table 8, fig. 9). Mean per-window AUROC falls from 0.694 (batch) to 0.428 (online), a drop of $\Delta = -0.266$, and the collapse is uniform across categories: DRIFT 0.681→0.448, OOD 0.766→0.479, and STABLE 0.628→0.349. The threshold-aware metrics move the same way — mean AUPR 0.698→0.540 and

Table 8: Batch (fit-once) vs. online incremental-covariance Mahalanobis under a true streaming protocol: mean per-window AUROC on the temporally-ordered TSB-StreamingAD-U evaluation stream. Δ = online − batch.

Category	n	Batch AUROC	Online AUROC	Δ
DRIFT	10	0.681	0.448	-0.234
OOD	10	0.766	0.479	-0.287
STABLE	9	0.628	0.349	-0.279
Overall	29	0.694	0.428	-0.266

FPR@95 worsens from 0.683 to 0.875. The effect is not an averaging artifact of a few datasets: across the 29 datasets the online variant *wins on* 1*, ties on* 2*, and loses on* 26 (per-dataset AUROC, tolerance ± 0.005).

The mechanism is intrinsic to unsupervised self-updating. Because the online covariance ingests *every* window it scores — including the anomalous Source-2 windows, which carry no label to exclude them — the running ID Gaussian gradually absorbs the shifted regime into its own support. As the covariance inflates along the shift directions, the Mahalanobis distance of subsequent anomalous windows shrinks toward the ID range, and the very separability the fit-once estimator preserves is eroded window by window. Self-updating thus defeats the purpose of the detector: the anomalies it is meant to flag are the same samples that contaminate its model of normality.

Two coverage caveats apply. The reported cells are single-seed (seed 42) and cover 10 files per category (29 datasets after the ordered-stream filter), so this is a representative-coverage probe rather than a full-corpus sweep; a larger 20-files-per-category run was OOM-killed before completion and is not reported. The direction and magnitude of the effect are nonetheless unambiguous (26 of 29 losses, $\Delta = -0.266$), and they carry a concrete deployment implication: the "deploy Mahalanobis" recommendation holds specifically for the *fit-once batch* estimator, and a naive self-updating online covariance is *not* a safe substitute for it.

7 Discussion

The benchmark yields a constructive result, not merely a cautionary one. It identifies a single detector family that works, explains why it works, quantifies how cheaply it works, and turns the fidelity audit that made the ranking trustworthy into a reusable methodology. We open with the takeaways, then develop the positive findings before analysing, in depth, why the confidence-based family fails.

7.1 Key findings and takeaways

- **There is a clear winner.** On the combined 527-dataset benchmark, detectors that model the in-distribution feature manifold occupy the entire top tier; Mahalanobis distance (0.826) and DFM-PCA (0.812) lead decisively across both univariate and multivariate families.
- **A concrete recommendation.** For streaming time-series OOD detection, *model the geometry and density of the in-distribution feature manifold* (Mahalanobis or DFM-PCA) rather than trusting the classifier's confidence; pair the detector with global normalisation for amplitude/level shifts.

- **The winners are cheap and self-contained.** They attain top accuracy at ∼1 ms to 4 ms per window (Mahalanobis 1.9 ms, DEEDEE 1.0 ms), require no auxiliary outlier data and no retraining, and are Pareto-optimal in accuracy versus latency.
- **Dependable across regimes.** The feature-manifold family is strong on the STABLE specificity controls (Mahalanobis 0.855) and the subtle DRIFT regime (0.858), so it neither misses drift nor over-flags near-stationary streams.
- **The audit is a first-class contribution.** CatSight, InvAD, DEEDEE, and even Mahalanobis reach the top tier *only* after principled, paper-checked corrections; the verify-before-experiment gate is a transferable methodology that most baseline studies omit.
- **A streaming extension of prior work.** These results independently confirm and extend the recommendation of TS-OOD [Gungor et al., 2025] from the static, same-dataset semantic-shift setting to streaming, window-level distributional shift, across 527 datasets and 17 detectors.
- **A disciplined cautionary result.** Every post-hoc softmax, logit, activation, and gradient detector falls *below chance* with systematically inverted scores; this is a genuine, audited failure of the confidence premise, not an implementation artifact.

7.2 What works, and why: the feature-manifold family

The central positive result is that detectors which model the in-distribution *feature* manifold dominate, and the mechanism is clean and general. These methods characterise the geometry and density of ID pre-logit features directly — Mahalanobis by a tied-covariance class-conditional Gaussian, DFM-PCA by per-class principal-component reconstruction error, DiMMAD by a continuous-metric distance ensemble — and score a test window by how far it lies from that learned manifold. Because they never consult the overconfident output head, they are structurally immune to the overconfidence pathology that sinks the post-hoc family (section 7.6): even when the classifier saturates on a far-out window, the feature vector still records the distributional gap, and distance/density in that space grows monotonically with the shift. The normalisation dichotomy of fig. 3 is the direct fingerprint of this mechanism: under global normalisation the second source retains its amplitude and level gap, so S2 windows sit many standard deviations outside the ID feature cloud, and the feature methods separate them cleanly (Mahalanobis 0.867, DFM-PCA 0.863 global). The practical virtues compound the accuracy advantage. The winners need no auxiliary outlier corpus and no retraining — they are fit on ID features alone — and they are among the cheapest detectors in the benchmark (section 6.5), so the accuracy–latency tradeoff that usually forces a compromise simply does not arise. For a practitioner the recommendation is therefore unambiguous and low-risk: default to Mahalanobis distance or DFM-PCA. One deployment refinement matters, established by our online-streaming test (section 6.8.5): the recommendation is to fit the Mahalanobis covariance *once* on the clean in-distribution warm-up and freeze it, *not* to let it self-update online, since an unsupervised incremental covariance absorbs anomalous windows into the running ID model and collapses the detector's separation (ΔAUROC $= -0.266$, losing on 26 of 29 datasets).

7.3 Category-level strengths as deployment guidance

The per-category breakdown (fig. 1) converts the winning family's robustness into actionable guidance. The feature-manifold detectors are strongest precisely where deployment demands reliability:

on the STABLE specificity controls, where a well-calibrated detector must *not* raise alarms, Mahalanobis (0.855), DFM-PCA (0.849), and DiMMAD (0.801) all maintain strong false-positive control; and on the subtle near-distribution DRIFT regime, where a gradual shift is easiest to miss, Mahalanobis reaches 0.858 and DFM-PCA 0.825. The strong-shift OOD category is the family's (relatively) weakest, around 0.76 for the two leaders — still comfortably above chance, but a useful reminder that a cleanly different S2 regime can push representations into a less-calibrated region of the tied-covariance model. The deployment reading is favourable and specific: use the feature-manifold family as a default; expect its highest confidence on stable and drifting streams; and, where the deployment is dominated by abrupt regime changes, consider ensembling the tied-covariance Mahalanobis with the reconstruction-based DFM-PCA, whose principal-subspace view is complementary.

7.4 The fidelity audit and its corrections as a methodology

A defining methodological contribution of this thesis is that implementation faithfulness is treated as a *measured precondition* rather than an assumption. We audited every candidate detector against its source paper and, where a repository exists, its official code (appendix A). All 24 audited candidates are legitimate OOD methods; 17 were faithful or admitted a validated, paper-faithful correction and are evaluated under the unified protocol, while the remaining 7 require an adapted, method-appropriate protocol and are evaluated separately for fairness (table 4). This is not bookkeeping: several of the top-tier results exist *only because* of the audit. CatSight (0.735) reaches the top tier after correcting an asserted orientation on its common-spatial-pattern distance; InvAD (0.740) is competitive once its affine-coupling invertibility is recognised to reduce the score to a feature-space statistic; DEEDEE (0.700) required its adjacent-dimension variant to be labelled and validated; and even Mahalanobis reaches the sole top position only after its within-class-scatter (tied-covariance) estimation was corrected. Most baseline studies never audit their baselines; by converting faithfulness from an unstated assumption into a per-method verdict with a synthetic-validation gate, we obtain a ranking whose top entries are trustworthy and whose failures are attributable. The audit is thus a reusable protocol — verify, correct where a paper-faithful fix exists, exclude where it does not, and report the ablation — as much as a set of results.

7.5 Honest corrections to the prior write-up

Reporting how the audit changed the headline is itself a result; four corrections stand out, each traceable to a specific per-method fix (appendix A).

(a) Mahalanobis is the sole top method. An earlier write-up reported a tie at the top between Mahalanobis and DriftLens. The audit established that the per-sample DriftLens implementation is a PCA-whitened Mahalanobis detector correlating at $\rho \approx 0.999$ with Mahalanobis proper — the same method under another name — and that the official DriftLens score is batch/window-level Fréchet drift, which the shuffled per-window protocol cannot realise. With DriftLens evaluated under its native batch-level arm rather than as a per-window duplicate, and the tied-covariance estimation corrected, Mahalanobis (0.826) stands alone ahead of DFM-PCA (0.812); the earlier tie was a relabelling artifact, not two independent methods converging.

(b) SRS falls below chance once restored to its true form. A prior write-up promoted SRS as the faithful, mechanistically distinct time-series method and reported it near the top. Restored to its true seasonal-ratio form — STL decomposition plus conditional-VAE scoring on the raw series

— SRS falls to 0.352, below the chance line and behind even DiffAD (0.531). The seasonal-ratio assumption presumes a stable seasonal template learned from the ID regime, which the abrupt two-source boundary violates; the one genuinely distinct time-series-specific method does not generalise to streaming shift. This is the most consequential reversal, and we report it plainly.

(c) CatSight, InvAD, and DEEDEE rank high only after fixes. These three now sit in the top tier (0.735, 0.740 and 0.700) but owe their standing to orientation, reduction, and variant corrections rather than to the mechanisms their names advertise; each is labelled an adaptation in appendix A. Reporting them under their original names without these caveats would overstate the fidelity of the result, so we do not.

(d) Faithful DICE (energy) fails with the post-hoc family. The corrected, paper-faithful DICE — signed top-k contribution sum followed by an energy score, matching Sun and Li [2022] — does not escape the collapse: it scores 0.313, squarely inside the inverted cluster alongside MSP (0.370), SCALE (0.308), ODIN (0.303), energy (0.301), and GradNorm (0.294). This is the crucial control on the audit: the post-hoc failure is *not* an artifact of buggy code. Even repaired to match its paper, the family inverts, because its defining assumption breaks under streaming shift.

7.6 Why post-hoc confidence fails below chance

The failure of the confidence-based family is deep enough to warrant a dedicated analysis, because "below chance" is a far stronger statement than "ineffective." We develop seven interlocking mechanisms.

1. Below chance means systematically inverted, not random. A detector at AUROC ≈ 0.5 is uninformative; the post-hoc cluster sits at 0.294 to 0.370, well *below* chance. By the probabilistic reading of AUROC, an in-distribution window receives a *higher* OOD score than a genuinely shifted window the majority of the time. These detectors are therefore reliably *wrong*, not noisy — there is real, exploitable signal, but it points the wrong way. A detector that is consistently inverted is more dangerous in deployment than one that is merely uninformative, because its confident answers are anti-correlated with the truth.

2. The overconfidence dichotomy. The inversion has a well-documented root: deep networks are frequently *more* confident on shifted or far-out inputs than on in-distribution ones. A deep ReLU network is provably free to be arbitrarily confident far from the training manifold [Hein et al., 2019, Nguyen et al., 2015]. When an S2 window arrives many standard deviations outside the ID region, the network can emit a near-maximal softmax probability; MSP, ODIN, energy, and ReAct then read that saturated confidence as strong evidence of *in*-distribution, assigning the OOD window a low anomaly score and the genuinely ID window a comparatively higher one. The confidence signal is not weak; it is pointed backwards.

3. The backbone is trained on temporal pseudo-classes. The pathology is sharpened by how the backbone is trained. There are no semantic labels in a streaming series, so the backbone is trained with cross-entropy over K *temporal pseudo-classes* formed by binning the ID training windows into arbitrary contiguous segments. The resulting confidence measures which temporal bin a window resembles, a quantity with no principled relationship to true distributional novelty. An OOD window can be assigned confidently to some pseudo-class simply because it happens to align with one arbitrary bin; confidence is decoupled from, and empirically anti-correlated with, genuine novelty.

4. Distribution shift is not semantic novelty. The post-hoc methods were designed for image classification, where OOD means a *new semantic class* and low confidence is a reasonable proxy for "none of my classes." Streaming distributional shift is a covariate shift within a label-free stream: there is no semantic class to be uncertain about. The premise that low confidence signals novelty simply does not transfer to this setting. Our result confirms and extends Gungor et al. [2025], who found post-hoc scores weak under static semantic shift, to the harder streaming regime where they do not merely weaken but invert.

5. A shared broken signal explains the tight cluster. MSP, ODIN, energy, ReAct, SCALE, DICE, and GradNorm all read the same over-confident logits (or their gradients), differing only in post-processing. Since the underlying signal is the same inverted confidence, they fail *together* and cluster within a narrow 0.08-wide band (0.294 to 0.370). This co-location is itself evidence for the mechanism: independent weaknesses would scatter, whereas a shared broken input produces a tight, correlated failure. By the same logic the feature-manifold family, which reads a different and intact signal, separates cleanly and wins together.

6. Global normalisation amplifies the inversion. The normalisation dichotomy (fig. 3) shows the mechanism operating as a dial. Under global normalisation the post-hoc cluster is most inverted (energy 0.214, GradNorm 0.213, ODIN 0.218), because the preserved out-of-range magnitudes push the logits to their most saturated, most overconfident extreme; under per-series normalisation, which rescales away the level gap, the same detectors recover toward — but never reach — chance (energy 0.452, GradNorm 0.433). The very preprocessing that most helps the feature methods most hurts the confidence methods, because a larger level gap is simultaneously cleaner geometry and stronger overconfidence.

7. This is not an orientation bug, and we do not flip the sign. It would be tempting to "fix" an inverted detector by negating its score. We do not, and the reason is a matter of scientific integrity. The Pass-1 audit verified that MSP and ODIN are faithful implementations with the *correct*, paper-defined orientation (appendix A); their inversion is the method's own behaviour, not a coding error. Negating the score to recover AUROC would be post-hoc orientation-fitting — choosing the sign after seeing the labels — which is precisely the malpractice the audit was built to catch (we document its danger in the regime-dependent DiverseMix correction, whose principled orientation reverses between synthetic and real data). The honest conclusion is that the method's own defined orientation genuinely fails on streaming shift; reporting the true, below-chance number is the point, not a problem to be papered over.

The separation is statistically significant and the inversion is systematic. Two extended analyses (section 6.8) put the headline on a firmer footing. First, a Nemenyi post-hoc test with a critical-difference diagram (fig. 5, table 6) shows that the feature–post-hoc gap is not merely an omnibus Friedman effect: each of the five top-ranked feature-manifold detectors significantly outranks *every* post-hoc softmax, logit, and gradient detector (all pairwise $p < 1 \times 10^{-15}$), and the ∼3-rank gap between the families is roughly three times the critical difference (CD = 1.016). Second, the orientation-stability analysis (table 5) shows the sub-chance means are not scattered near-misses but a *systematic* inversion — the post-hoc detectors score below 0.5 on 58 % to 73 %

of datasets — and that the inversion is *regime-dependent*, roughly 20 % more prevalent under global than per-series normalisation (e.g. energy 79.3 % versus 52.8 %). Both results reinforce the mechanistic story of section 7.6: a shared, sign-inverted confidence signal, amplified exactly by the preprocessing that most helps the feature geometry. A global sign flip would recover most of the lost AUROC (table 5), which we deliberately do not perform, for the integrity reasons above.

7.7 A single winning principle

A publication-relevant subtlety is that several detectors marketed as distinct — DFM-PCA (PCA reconstruction), DiMMAD (distance ensemble), the adapted-protocol DriftLens (PCA-Mahalanobis) and TD-IVDM (KDE density), and Mahalanobis itself — are, in their per-sample implementations, all variants of distance or density estimation in the frozen feature space. The impression that "a diverse set of methods wins" is therefore partly an artifact of relabelling: the robust signal is feature-space distance/density, surfacing under several names. This sharpens rather than weakens the headline — it identifies a single winning principle rather than a coincidence of unrelated methods, and it is the reason two of the seven adapted-protocol methods (table 4) reduce to feature-space duplicates under a per-window arm rather than adding genuinely new detectors to the unified leaderboard. For the practitioner it is reassuring: the recommendation reduces to one idea, robustly instantiated.

7.8 Extending TS-OOD to the streaming regime

The study sits in the empirical-benchmark lineage of Gungor et al. [2025] and extends it along three axes. Where TS-OOD evaluated eight detectors on thirteen datasets under a static, same-dataset semantic-shift protocol, we evaluate 17 audited detectors on 527 datasets under a leakage-free, window-level streaming-shift protocol spanning univariate and multivariate families. The core recommendation — that deep feature modeling is the most promising direction for time-series OOD detection — not only survives this much harder setting but is stated more strongly: under streaming shift the post-hoc alternatives do not merely lag, they invert. The breadth of coverage (527 datasets, both modalities, three difficulty categories, two normalisation schemes) and the strongly significant Friedman result together make this, to our knowledge, the most comprehensive fair evaluation of OOD detectors on streaming time series to date, and a positive, constructive corroboration of the feature-modeling thesis.

7.9 Threats to validity and limitations

We state the scope honestly; these are scoping decisions that define the natural next experiments rather than undermining the present conclusions, which rest on the very strongly significant Friedman result and the consistent rank ordering across both families, categories, and normalisation schemes.

1. **CPU-only.** All experiments run on CPU; the inference-time comparisons (fig. 4) are CPU latencies and would change in absolute terms, though not in ranking, on accelerated hardware.

2. **Friedman complete-case caveat.** The 17-detector complete-case Friedman pool is $N = 250$ univariate datasets, because SRS is univariate-seasonal and is not run on the multivariate split. A combined-family test over the full corpus (16 detectors excluding SRS, $N = 515$ univariate and multivariate) is equally highly significant ($\chi^2 = 2336.59$, $p < 1 \times 10^{-16}$), so the conclusion holds across both splits; only an SRS-inclusive combined test is out of scope.

3. **Small Source-1 training windows.** The source-boundary split trains only on the pre-boundary Source-1 region, typically a small leading fraction of each file; the resulting ID pool is modest, which particularly constrains density estimators fit on few windows.

4. **Adapted-protocol methods run under separate protocols.** The seven adapted-protocol detectors (table 4) are reported in table 3 under their own method-appropriate arms on both TSB-U and -M; even so, none rises clearly above chance (per-arm AUROC 0.449 to 0.696). A full-scale study under the protocols they *are* designed for (ordered streams, auxiliary outlier corpora, batch-level drift) remains future work. Evaluating them separately is the honest choice for fairness, but it means the comparison to the unified-protocol set is indicative, not like-for-like.

5. **Single seed.** All splits and initialisations use seed 42; a multi-seed study would attach confidence intervals to every AUROC.

7.10 Promise and future directions

The limitations map directly onto concrete next steps, two of which this work has already resolved. The orientation stability of the energy-based scores is settled here rather than deferred: the orientation-stability analysis (section 6.8.1) shows when the sub-chance scores can and cannot be trusted without post-hoc sign-fitting. Likewise, the online deployment of the Pareto-optimal feature-manifold detectors is no longer a promissory note but a measured — and negative — result: an incremental-covariance Mahalanobis run under a true streaming protocol *underperforms* the fit-once batch detector (ΔAUROC $= -0.266$, section 6.8.5), because unsupervised per-window updates absorb the anomalous windows into the running in-distribution model. Of the remaining directions, two are genuinely open. A multi-seed study would replace point estimates with confidence intervals; and re-training the backbone with a multi-positive contrastive loss is expected, per Gungor et al. [2025], to lift the activation-based detectors specifically, testing whether the post-hoc collapse is intrinsic to the confidence premise or an artifact of the pseudo-class backbone. A full SRS multivariate sweep, finally, is deferred as low-value rather than a gap: the Friedman complete-case gap is already closed by the 16-detector ex-SRS full-coverage test ($\chi^2 = 2336.59$, $N = 515$), and a feasibility probe confirms SRS runs on the multivariate split only at near-chance AUROC and with an STL cost scaling in the training-window count, prohibitive at full scale. For practitioners deploying OOD detection on streaming time series today, the recommendation is already concrete and positive: *default to fit-once batch Mahalanobis distance or DFM-PCA*, prefer global normalisation for amplitude/level shifts, and do not rely on post-hoc softmax, logit, or gradient scores, which invert under exactly the far-out inputs they are meant to flag.

8 Conclusions

We set out to test whether the central recommendation of TS-OOD [Gungor et al., 2025] — that deep feature modeling beats post-hoc softmax scoring for time-series OOD detection — survives the move from static semantic shift to streaming distributional shift, and whether it holds across a detector pool far broader than the original eight. After auditing twenty-four candidate implementations for faithfulness and evaluating seventeen on the two target families TSB-StreamingAD-U and TSB-StreamingAD-M across 527 datasets and 8668 runs, the answer is clear. Distance- and density-based detectors that model the in-distribution feature manifold — Mahalanobis distance (0.826) and DFM-PCA (0.812) foremost — lead the ranking, while every post-hoc softmax, logit, and gradient detector (and the seasonal SRS) falls below the 0.5 chance line, several with inverted scores; the Friedman test confirms the ordering is highly significant ($\chi^2 = 2336.59$, $p < 1 \times 10^{-16}$). The same feature-space detectors are Pareto-optimal in accuracy versus inference cost, attaining top accuracy at roughly 1 ms to 4 ms per window.

Two findings sharpen the picture. First, the univariate family is modestly easier than the multivariate family for the feature cluster. Second, a faithfulness audit revealed pervasive implementation drift in the post-hoc family, yet even paper-faithful corrected variants remain below chance — the post-hoc failure is real, not a bug. A cautionary negative result accompanies this: the correct orientation of energy-based scores is regime-dependent, so a fix validated on synthetic data can reverse on real data.

For practitioners deploying OOD detection on streaming time series, the recommendation is concrete: *default to fit-once batch Mahalanobis distance or DFM-PCA*, choose the normalisation scheme according to the expected shift type (global for amplitude/level shifts, per-series for morphological shifts), and do not rely on post-hoc softmax or logit scores, which invert under exactly the far-out inputs they are meant to flag. The "fit-once" qualifier is load-bearing: our online-streaming test (section 6.8.5) shows that letting the Mahalanobis covariance self-update online, rather than fitting it once and freezing it, collapses the detector, so a naive incremental covariance is not a safe substitute.

This work also settles two questions we had previously flagged as open. First, the orientation stability of the sub-chance post-hoc scores is now a measured result rather than a conjecture: the extended analyses (section 6.8.1) show the inversion is systematic (post-hoc detectors below chance on 58 % to 73 % of datasets) and regime-dependent, roughly 20 % more prevalent under global than per-series normalisation. Second, we tested the online-deployment idea directly and report it as a negative result (section 6.8.5): an incremental-covariance Mahalanobis run under a true streaming protocol *underperforms* the fit-once batch detector (mean AUROC 0.428 vs 0.694, $\Delta = -0.266$, losing on 26 of 29 datasets), because unsupervised per-window updates absorb the anomalous windows into the running in-distribution model.

Two directions remain genuinely open. A multi-seed study would attach confidence intervals to every AUROC, replacing the single-seed point estimates. Re-training the backbone with a multi-positive contrastive (MPC) loss is expected to lift the activation-based detectors specifically, per Gungor et al. [2025], testing whether the post-hoc collapse is intrinsic to the confidence premise or an artifact of the pseudo-class backbone. A full SRS multivariate sweep, by contrast, is now deferred as low-value future work rather than a gap: the Friedman complete-case gap is already closed by the 16-detector ex-SRS full-coverage test ($\chi^2 = 2336.59$, $N = 515$), and a feasibility probe confirms that SRS does run on TSB-StreamingAD-M but at near-chance AUROC and with an STL cost that scales in the training-window count, making a full-scale sweep prohibitive for little expected information.

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A Implementation Fidelity Audit

Every detector implementation was audited against its published description and, where a repository exists, against its official code, before any experiment was run. All twenty-four audited candidates are legitimate OOD methods from the literature; seventeen were faithful or admitted a validated, paper-faithful correction and are evaluated under the unified streaming protocol, while the remaining seven require an adapted, method-appropriate protocol and are evaluated separately for fairness (table 4, table 3). Table 9 summarises the per-method fidelity verdict and evaluation protocol.

The corrected and adapted main-set variants. Several main-benchmark discrepancies admitted a paper-faithful correction, implemented and validated as `_enh` variants (section 7). Four required a CORRECTED fix. `gradnorm_enh` computes the L_1 norm of the last-layer-weight gradient of the KL-to-uniform objective and negates it (matching the published confidence orientation). `react_enh` computes energy after activation clipping. `scale_enh` applies activation scaling at the penultimate layer. Mahalanobis was corrected to a within-class-scatter (tied-covariance) estimate. A further group is ADAPTATION: `dice_enh` sums the signed top-k contributions and computes energy (faithful to Sun and Li [2022]); `dimmad_enh` drops the binary set metrics (Hamming, Jaccard, Dice) that are ill-defined on continuous features; and CatSight, InvAD, and M2N2 recover their faithful mechanism after orientation, reduction, and EMA-leak fixes respectively. None of these corrections lifts a post-hoc detector into competitiveness with the feature-space methods.

The adapted-protocol orientation cautionary result. The augmented-training method `diversemix_enh` (evaluated under its adapted protocol, table 3) negates the score to match the training objective; on the synthetic validation task this raised AUROC from 0.188 to 0.812, yet on real streaming data the same correction lowered mean AUROC from 0.633 to 0.367 — the regime-dependent orientation result discussed in the main text, and the reason we do not post-hoc sign-fit inverted detectors.

The feature-space relabelling. The audit also established that the per-sample implementations of DiMMAD (distance ensemble), DFM-PCA (PCA reconstruction), and Mahalanobis — together with the adapted-protocol methods DriftLens (PCA-Mahalanobis) and TD-IVDM (KDE density) — are all variants of distance/density estimation in the frozen feature space. The robust winning signal is therefore feature-space distance/density surfacing under several names, which reframes the headline result; it is also why DriftLens and TD-IVDM, whose native protocols are batch/window-level, are evaluated under their adapted arms rather than as duplicate per-window top-ranks in the unified leaderboard.

The systemic finding. Across the audit, numerous detectors contained a score negation or orientation choice rationalised post-hoc by an observation on small training data rather than following the principled score from the paper. In the unified-protocol set this affected GradNorm, DiffAD, CatSight, CODiT, and the M2N2 EMA path, each repaired to its paper-defined orientation before evaluation; in the adapted-protocol set it affected DiverseMix and AE-ADWIN-LSTM. Together with the absent OE/DivOE auxiliary-outlier training, GradNorm's wrong statistic, DiffAD's input-independent reverse process, and the feature-space relabelling above, this means an unaudited ranking would substantially reflect implementation artifacts. This is the central methodological finding and the rationale for the verify-before-experiment gate.

B ROC Curves and Score-Distribution Case Study

C Reproducibility

All experiments fix the random seed to 42 and run on CPU under Python 3.13 with PyTorch, NumPy, SciPy, and scikit-learn. The backbone is a 1-D ResNet trained for 40 epochs with cross-entropy over temporal pseudo-classes. The benchmark comprises 527 datasets (260 TSB-StreamingAD-U and 267 TSB-StreamingAD-M), 17 evaluated detectors, and 8668 completed runs. Per-run configurations and metrics are stored alongside the aggregate results matrix, and the tables and figures in this manuscript are regenerated directly from it. The reduced epoch budget is recorded explicitly as a scoping decision; SRS is univariate-seasonal and its multivariate run is pending, so it is evaluated on TSB-StreamingAD-U only; and the seven adapted-protocol detectors (table 4) are evaluated under their native arms and reported separately in table 3.

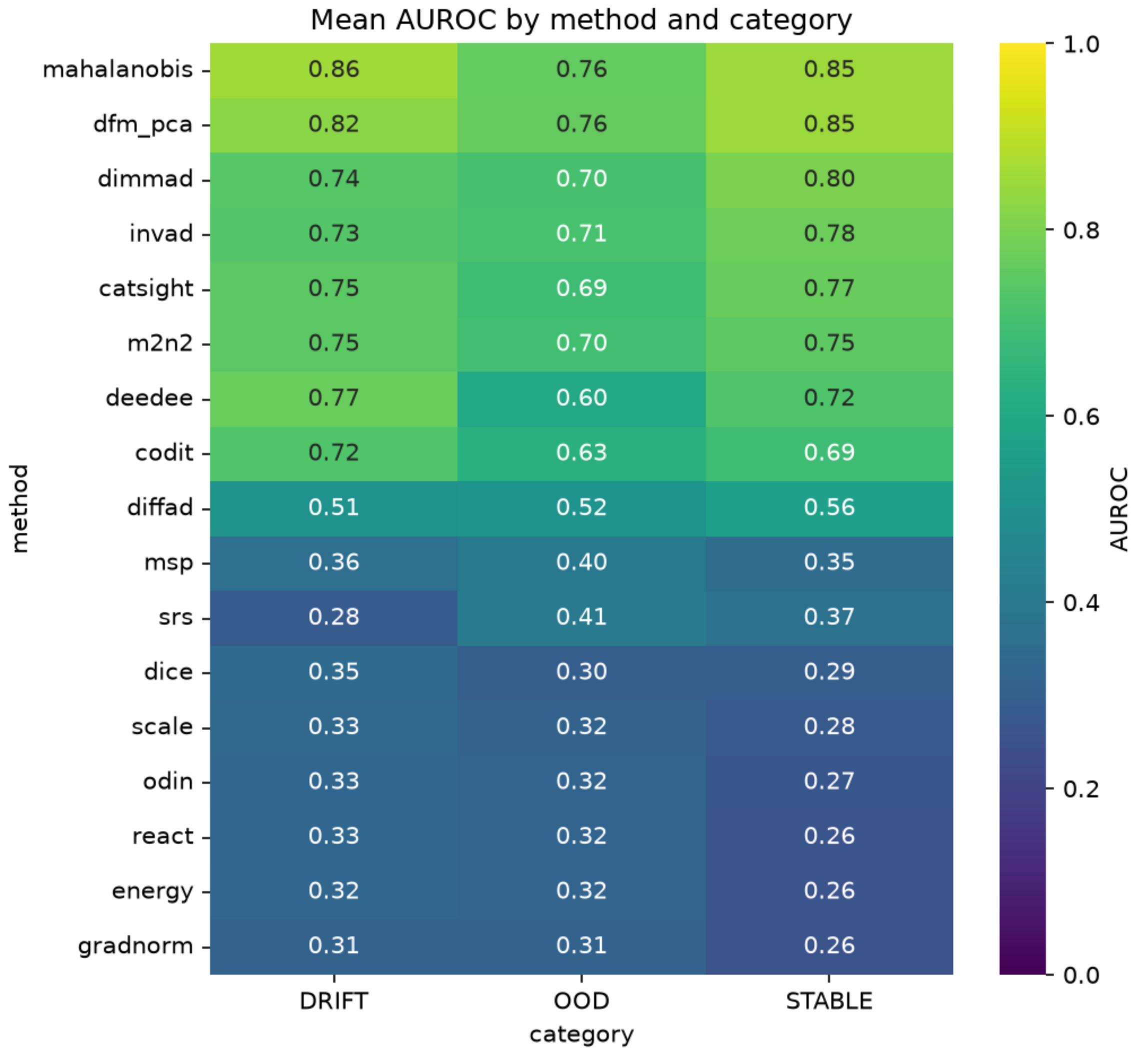

Figure 1: Mean AUROC of every detector (rows, ordered by overall combined mean) on the three TSB-StreamingAD categories DRIFT, OOD, and STABLE (columns). Brighter cells denote higher AUROC. The top rows — feature distance/density and feature-manifold detectors — are bright across all three categories, while the bottom rows — post-hoc softmax, logit, activation, and gradient detectors, together with the seasonal SRS — are uniformly dark, indicating scores at or below the 0.5 chance line. Among the leaders, the STABLE specificity controls and the subtle DRIFT regime are separated best (Mahalanobis 0.855 STABLE, 0.858 DRIFT; DFM-PCA 0.849 STABLE), while the post-hoc cluster is darkest on STABLE, where it over-flags the very control windows a calibrated detector should leave alone.

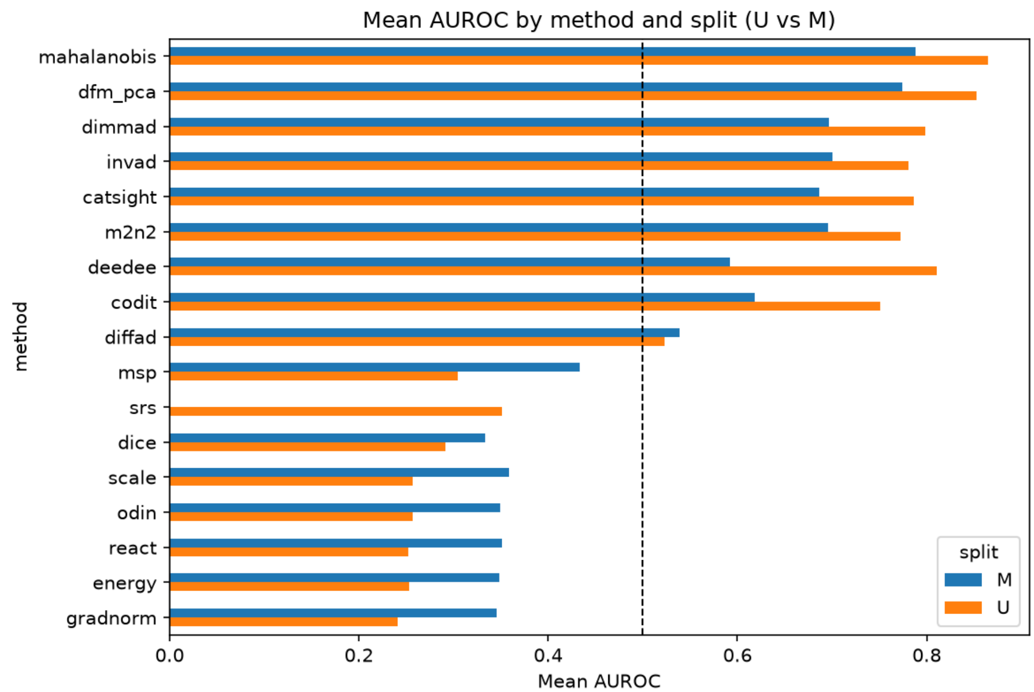

Figure 2: Mean AUROC per detector on TSB-StreamingAD-U versus TSB-StreamingAD-M. The feature distance/density and feature-manifold detectors (upper band) cluster tightly above 0.59 on both families, whereas the post-hoc softmax, logit, activation, and gradient detectors (lower band) fall below the 0.5 chance line on both, several with inverted scores. The two clusters are cleanly separated with no overlap; the post-hoc cluster is marginally less inverted on the multivariate family.

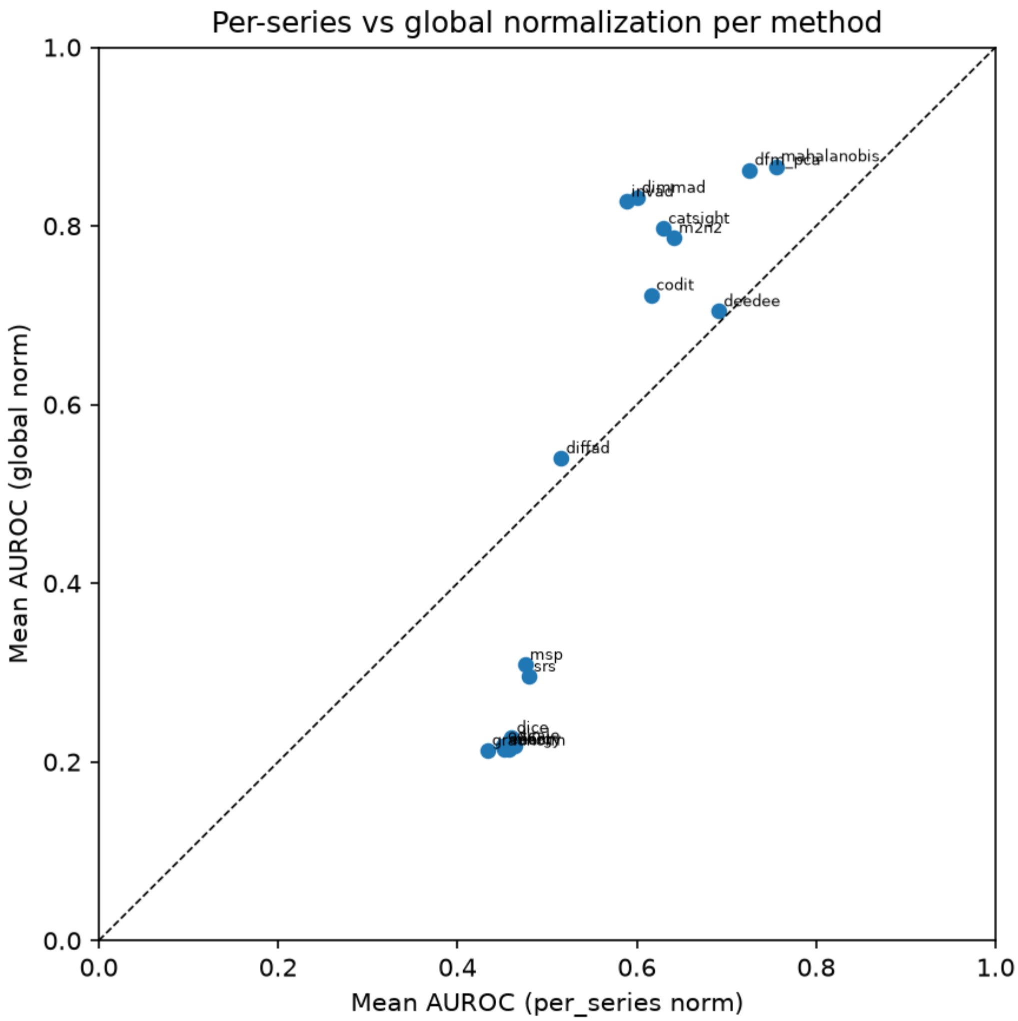

Figure 3: Mean AUROC under global (Source-1) normalisation versus per-series normalisation, per detector. Feature distance/density detectors (upper points) gain sharply under global normalisation; post-hoc softmax/logit/gradient detectors (lower points) remain inverted under both schemes and are slightly less inverted under per-series normalisation. The two clusters move in opposite directions — a dichotomy, not a shared preprocessing preference.

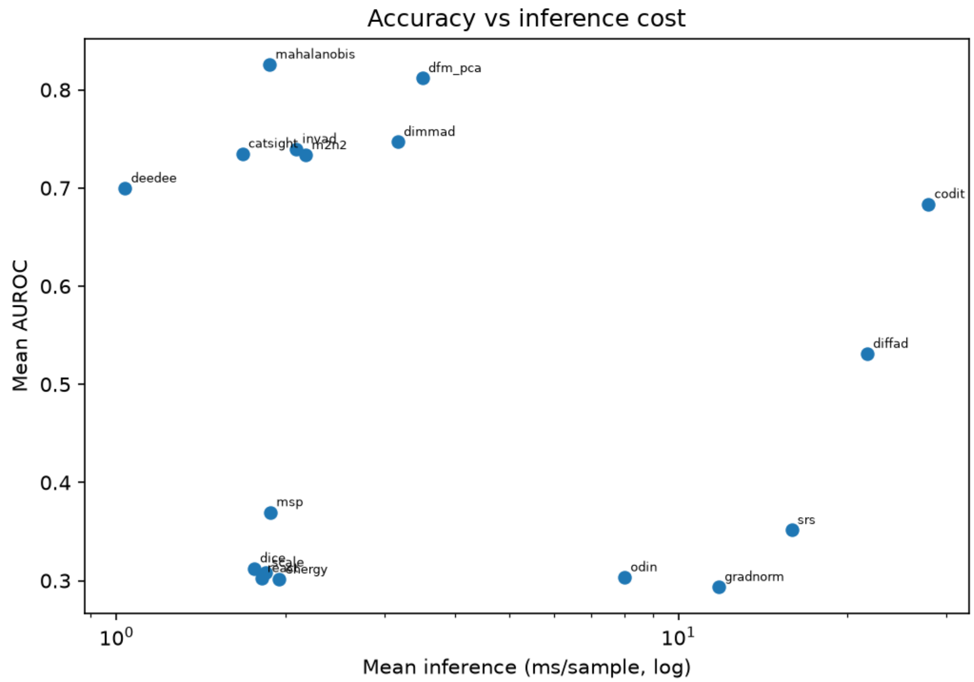

Figure 4: Accuracy versus inference cost (time axis on log scale). Each point is a detector; the upper-left region is best. Mahalanobis, DFM-PCA, DEEDEE, and CatSight hold the Pareto front, attaining the highest mean AUROC at roughly 1 ms to 4 ms per window, whereas CODiT (∼28 ms), DiffAD (∼22 ms), and SRS (∼16 ms) are far slower without a corresponding accuracy gain.

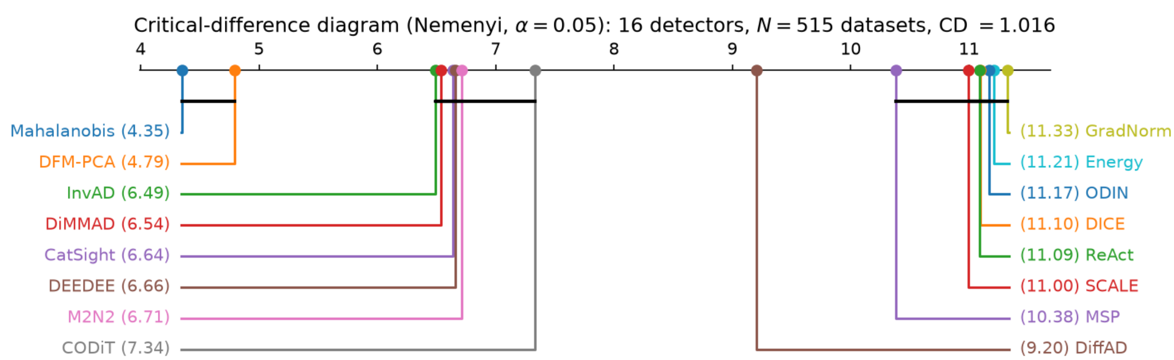

Figure 5: Critical-difference diagram (Nemenyi post-hoc, $\alpha = 0.05$) over the 16 detectors excluding SRS on the complete-case pool of $N = 515$ datasets. Detectors are positioned by mean rank (rank 1, best, at the top); a crossbar joins detectors whose mean ranks differ by less than the critical difference (CD = 1.016). The feature-manifold detectors (Mahalanobis 4.35, DFM-PCA 4.79, then the InvAD–M2N2 band) are cleanly separated from the post-hoc cluster (mean ranks 10.38 to 11.33), which is itself a statistical tie internally.

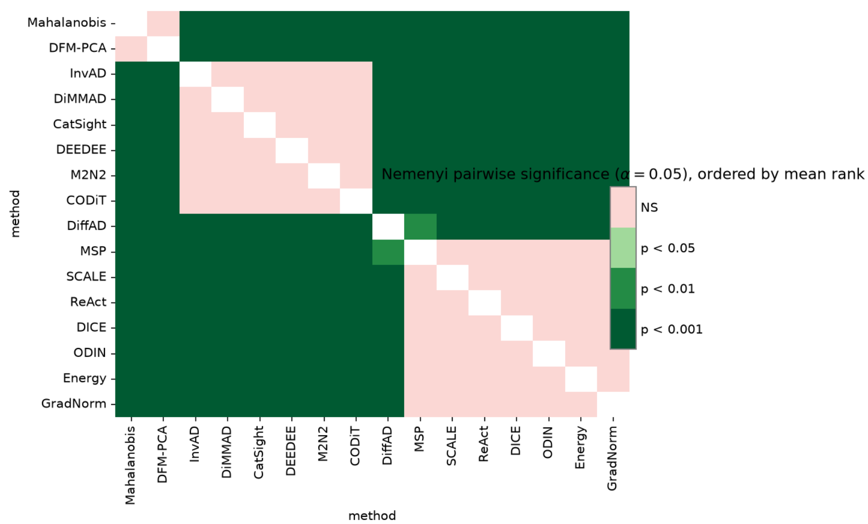

Figure 6: Nemenyi pairwise-significance sign-plot over the 16 detectors (ordered by mean rank). A coloured off-diagonal cell marks a significant pairwise difference at $\alpha = 0.05$; the dark block separating the feature-manifold detectors (upper-left) from the post-hoc cluster (lower-right) shows that every cross-family contrast is significant, while the post-hoc detectors are mutually indistinguishable.

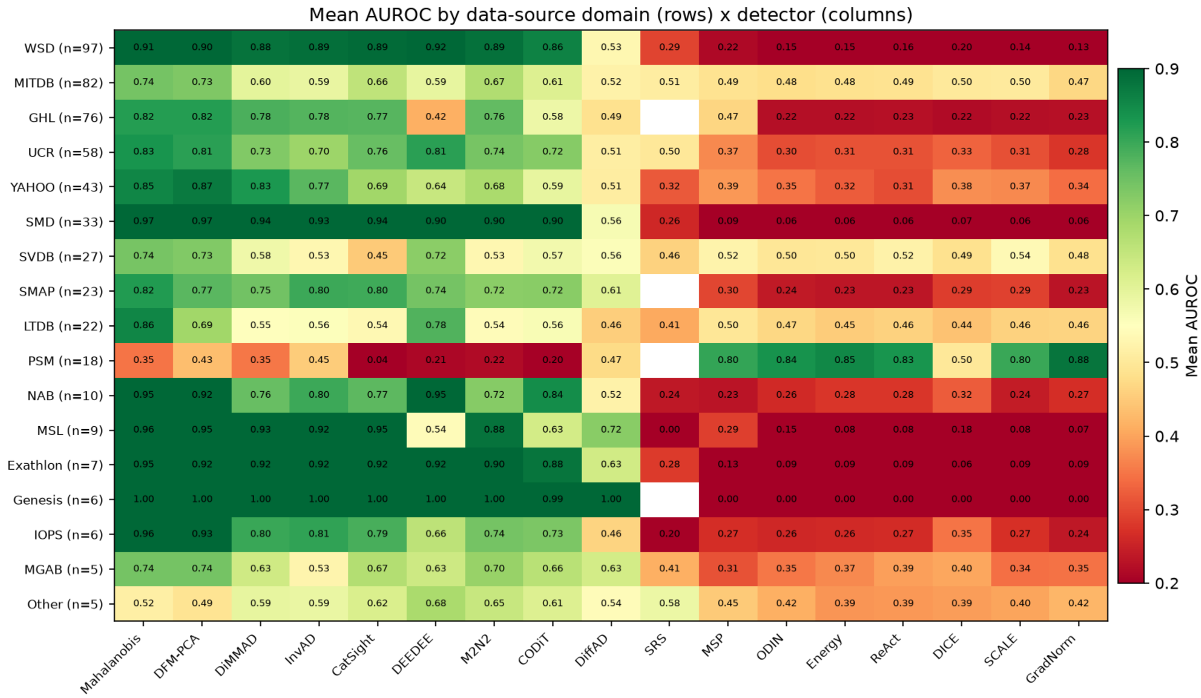

Figure 7: Mean AUROC by data-source domain (rows, with dataset counts) and detector (columns, grouped feature-manifold, then ts-specific, then post-hoc). Green marks strong separation, red marks inversion. The left (feature-manifold) block is green across almost every domain; the right (post-hoc) block is red across almost every domain. The PSM row inverts the pattern — red on the left, green on the right — the single domain where the post-hoc family outperforms the feature family.

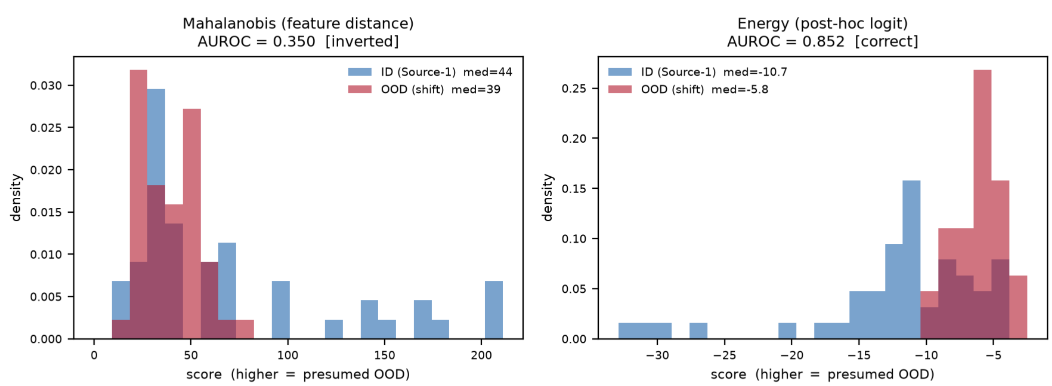

Figure 8: PSM reversal on the shared 96-window OOD test set (representative file `OOD_017_…_PSM`). *Left:* Mahalanobis feature distance — shifted (OOD) windows are nested *inside* and less dispersed than the in-distribution (ID) windows, so the distance score inverts (AUROC 0.350). *Right:* post-hoc Energy — the same shifted windows sit at higher energy (lower confidence) than ID windows, so the score ranks correctly (AUROC 0.852). Higher is presumed-OOD in both panels.

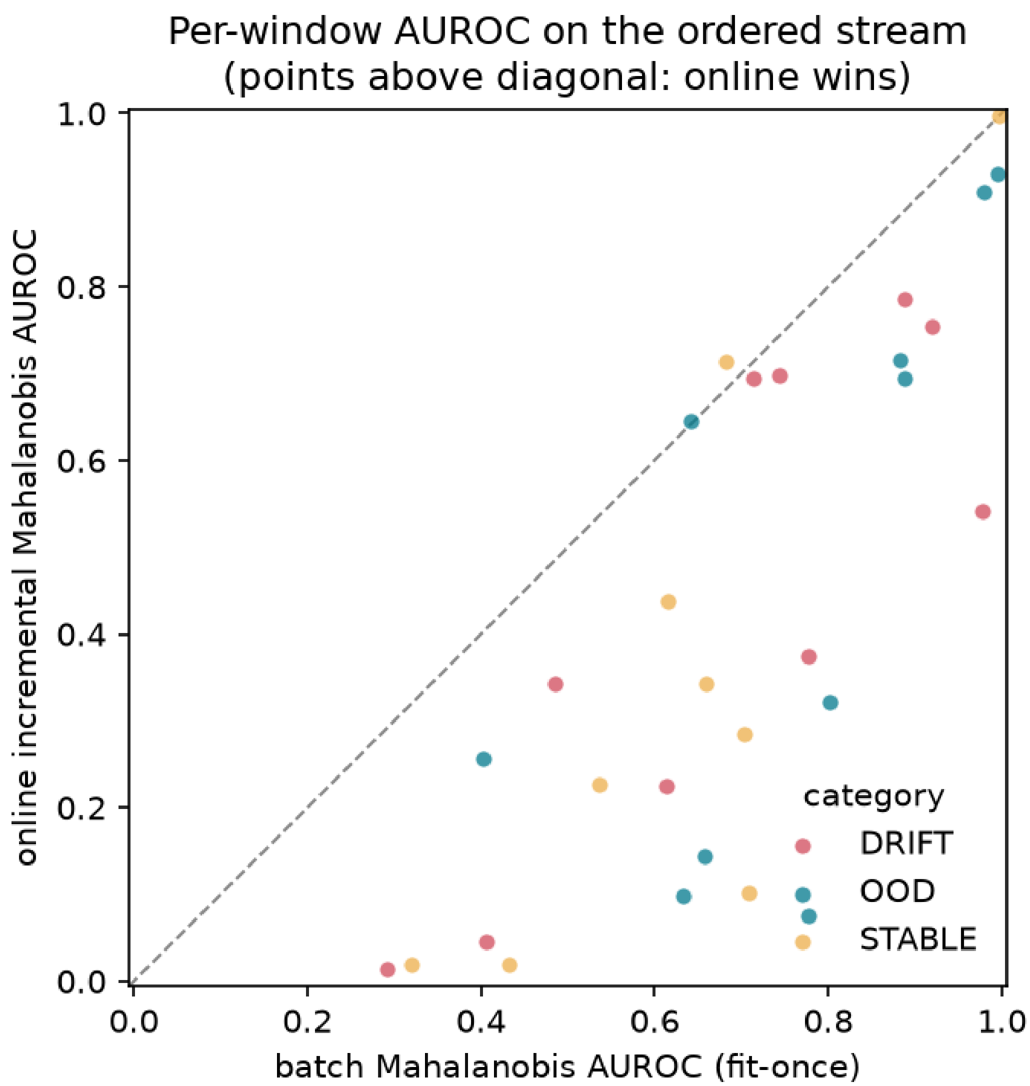

Figure 9: Batch (fit-once) vs. online incremental-covariance Mahalanobis on the temporally-ordered TSB-StreamingAD-U evaluation stream (seed 42, 29 datasets, online decay 0.999). The online self-updating covariance trails the fit-once batch detector on mean per-window AUROC in every category (overall $\Delta = -0.266$; win/tie/loss = 1, 2 and 26), because unsupervised per-window updates absorb anomalous windows into the running in-distribution Gaussian and collapse the Mahalanobis separation.

Table 9: Per-method fidelity verdict and evaluation protocol for the twenty-four audited candidate detectors. All twenty-four are legitimate OOD methods from the literature audited for fidelity; the final column indicates the *evaluation protocol*, not exclusion. The *fidelity verdict* records the audit outcome: FAITHFUL — the implementation matches the source paper (and official code where available) as-is; CORRECTED — a discrepancy that materially changed the OOD score was repaired with a validated, paper-faithful fix (an `_enh` variant, section 4); ADAPTATION — the faithful mechanism was recovered but relabelled or restricted to remain well-defined on the frozen-backbone protocol. The *evaluation* column states the protocol under which the detector is evaluated: *Unified* for the seventeen compatible with the shared fair-comparison streaming protocol, or *Adapted* for the seven whose defining mechanism requires a method-appropriate protocol (auxiliary-outlier training, an ordered stream, a retrained feature extractor, or batch-level scoring) and are reported separately (table 4, table 3). All five post-hoc/activation/gradient methods that a prior draft marked "not verified" (ReAct, DICE, SCALE, GradNorm) and DiffAD were either corrected or confirmed faithful and are evaluated under the unified protocol; the three augmented-training methods (Outlier Exposure, DivOE, DiverseMix) require an auxiliary-outlier training regime the unified protocol cannot provide, so they are evaluated under adapted arms.

Detector	Cluster	Fidelity verdict	Evaluation
Unified-protocol methods (17)			
MSP	post-hoc softmax	Faithful	Unified
ODIN	post-hoc softmax	Faithful	Unified
Energy (EBO)	post-hoc logit	Faithful	Unified
ReAct	activation shaping	Corrected (`react_enh`)	Unified
DICE	activation shaping	Faithful (`dice_enh`)	Unified
SCALE	activation shaping	Corrected (`scale_enh`)	Unified
GradNorm	gradient	Corrected (`gradnorm_enh`)	Unified
Mahalanobis (MDS)	feature distance	Corrected (within-class scatter)	Unified
DFM-PCA	feature density	Faithful	Unified
DiMMAD	feature distance	Adaptation (metric ensemble)	Unified
CatSight	concept drift	Adaptation (orientation fixed)	Unified
InvAD	ts-specific	Adaptation (reconstruction restored)	Unified
M2N2	test-time adaptation	Adaptation (EMA-leak fixed)	Unified
DEEDEE	ts-specific	Faithful (`deedee_fix`)	Unified
CODiT	ts-specific	Faithful (on-protocol)	Unified
SRS	ts-specific	Faithful (seasonal ratio)	Unified
DiffAD	reconstruction	Faithful (`diffad_fix`)	Unified
Adapted-protocol methods (7)			
Outlier Exposure	augmented training	Faithful rebuild	Adapted
DivOE	augmented training	Faithful rebuild	Adapted
DiverseMix	augmented training	Faithful rebuild	Adapted
DriftLens	drift/embedding	Adaptation (relabel)	Adapted
TD-IVDM	concept drift	Adaptation (relabel)	Adapted
AE-ADWIN-LSTM	concept drift	Faithful rebuild	Adapted
DIVERSIFY	ts-specific	Adaptation (relabel)	Adapted

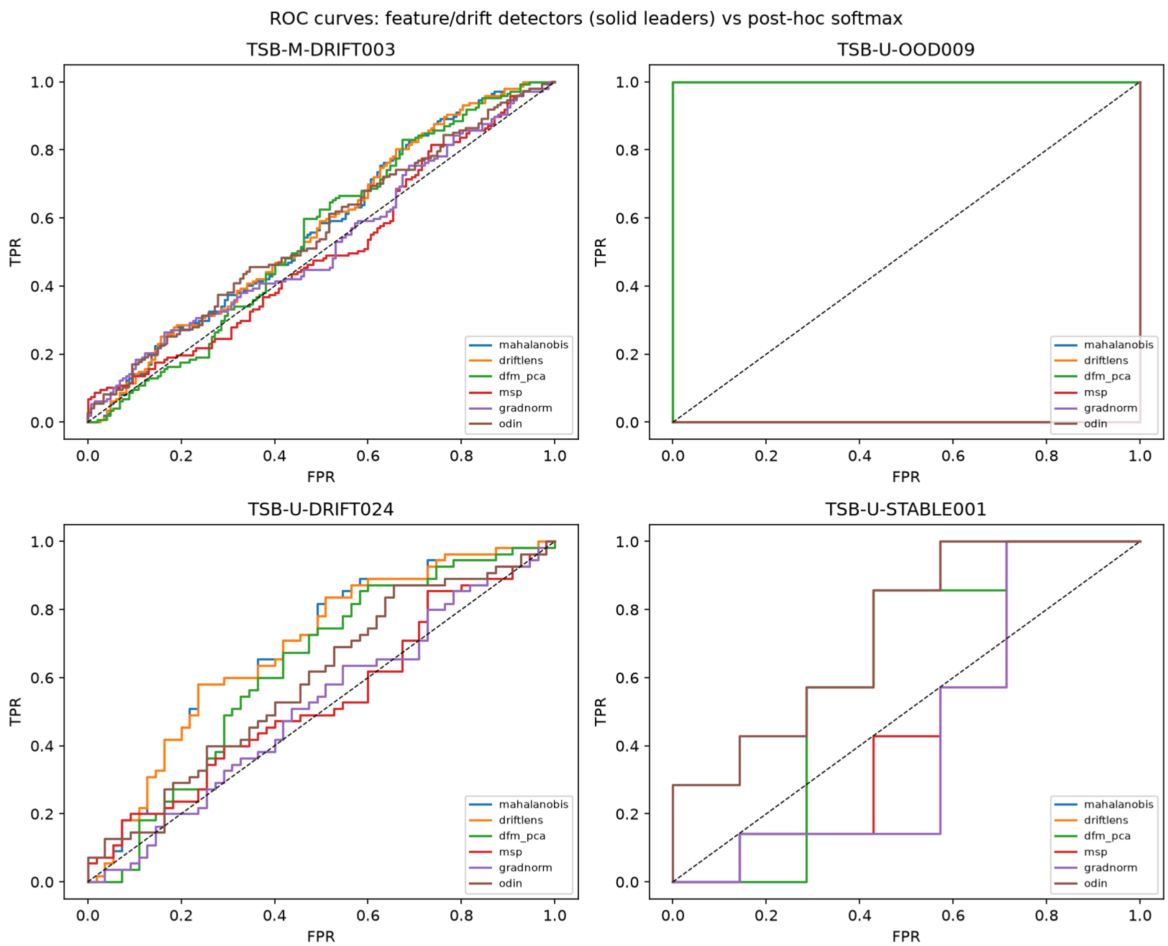

Figure 10: Receiver-operating-characteristic curves for three leading feature/drift detectors and three post-hoc softmax detectors on a representative global-normalised dataset. The softmax curves fall below the diagonal, the signature of score inversion, while the feature-space curves hug the top-left corner.

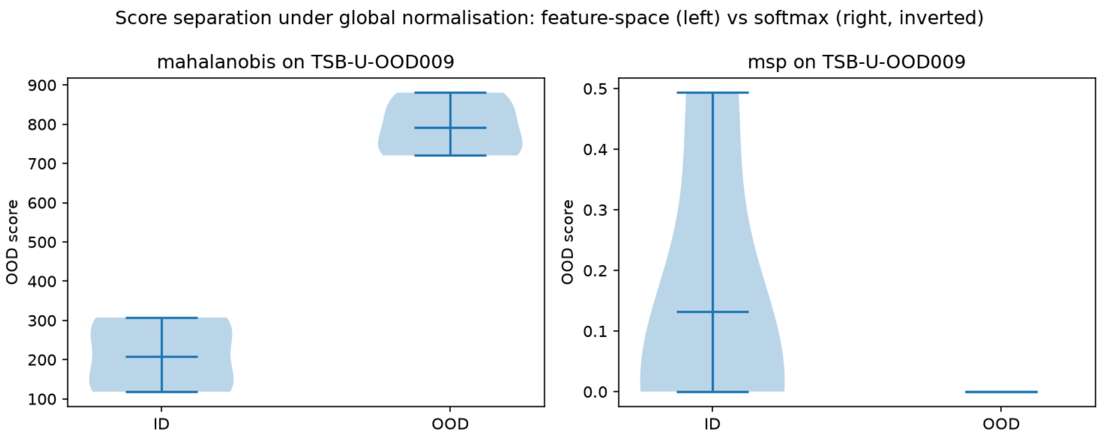

Figure 11: Distribution of out-of-distribution scores for in-distribution versus out-of-distribution test windows on a global-normalised dataset. The feature-space detector (left) assigns systematically higher scores to OOD windows; the softmax detector (right) assigns them *lower* scores, inverting the decision and producing an AUROC below chance — the mechanism analysed in the main text.